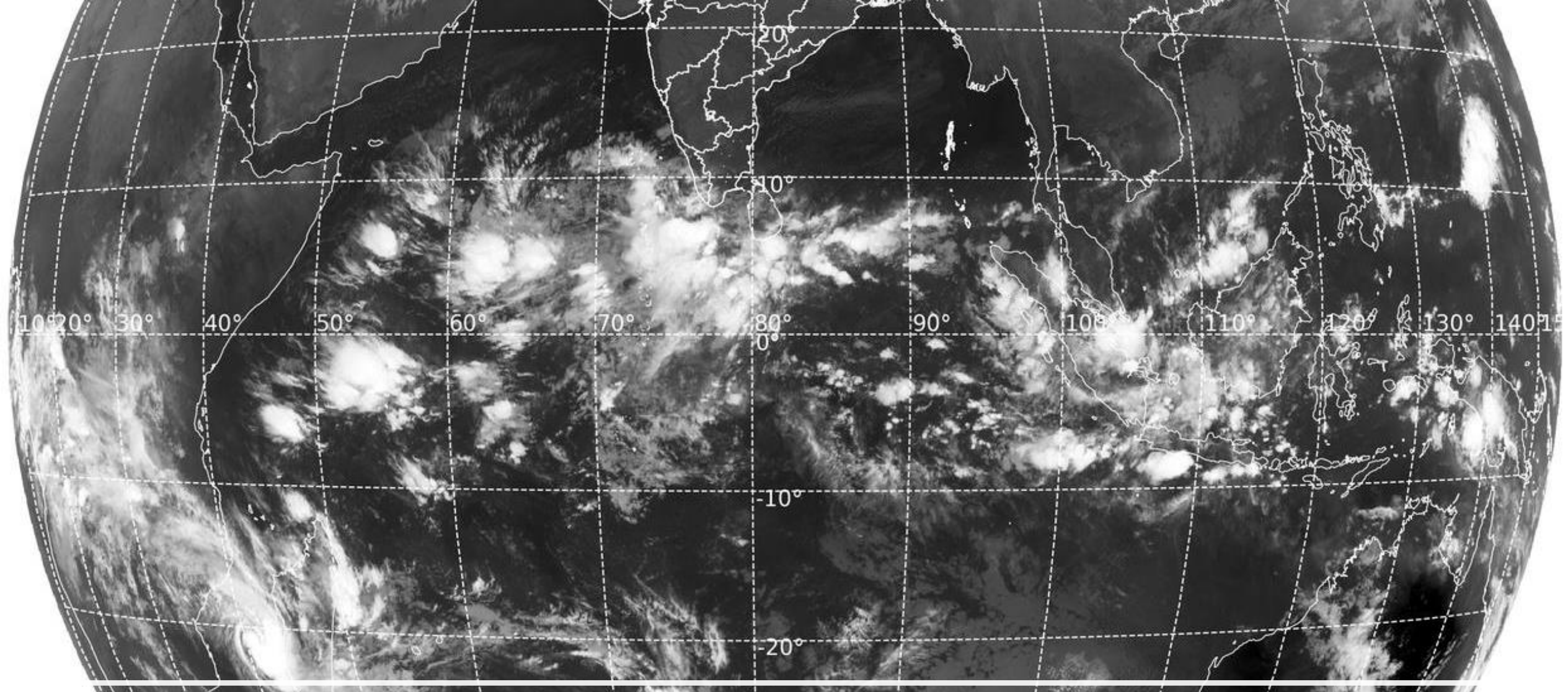


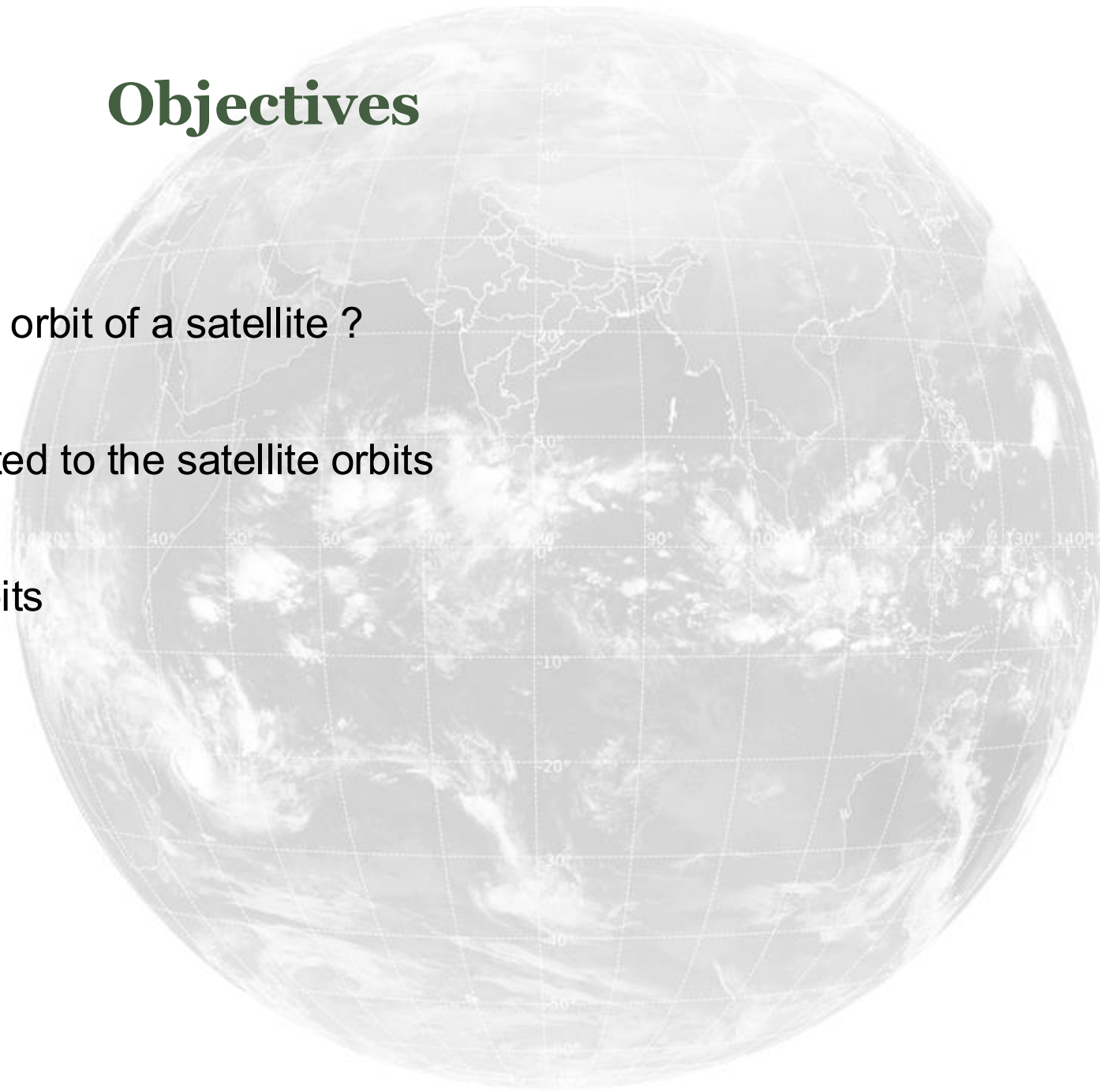


Remote Sensing and Geographic Information Science

Note: In this ppt, figures/text are adopted from various sources; this is only for class explanation purpose.

Objectives

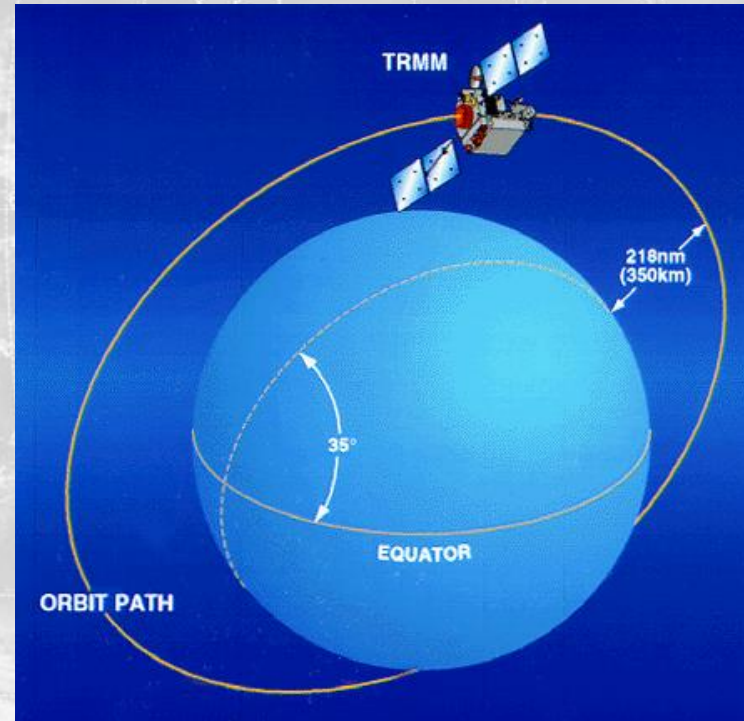
- What is meant by an orbit of a satellite ?
- Technical terms related to the satellite orbits
- Types of satellite orbits



Orbit of a Satellite

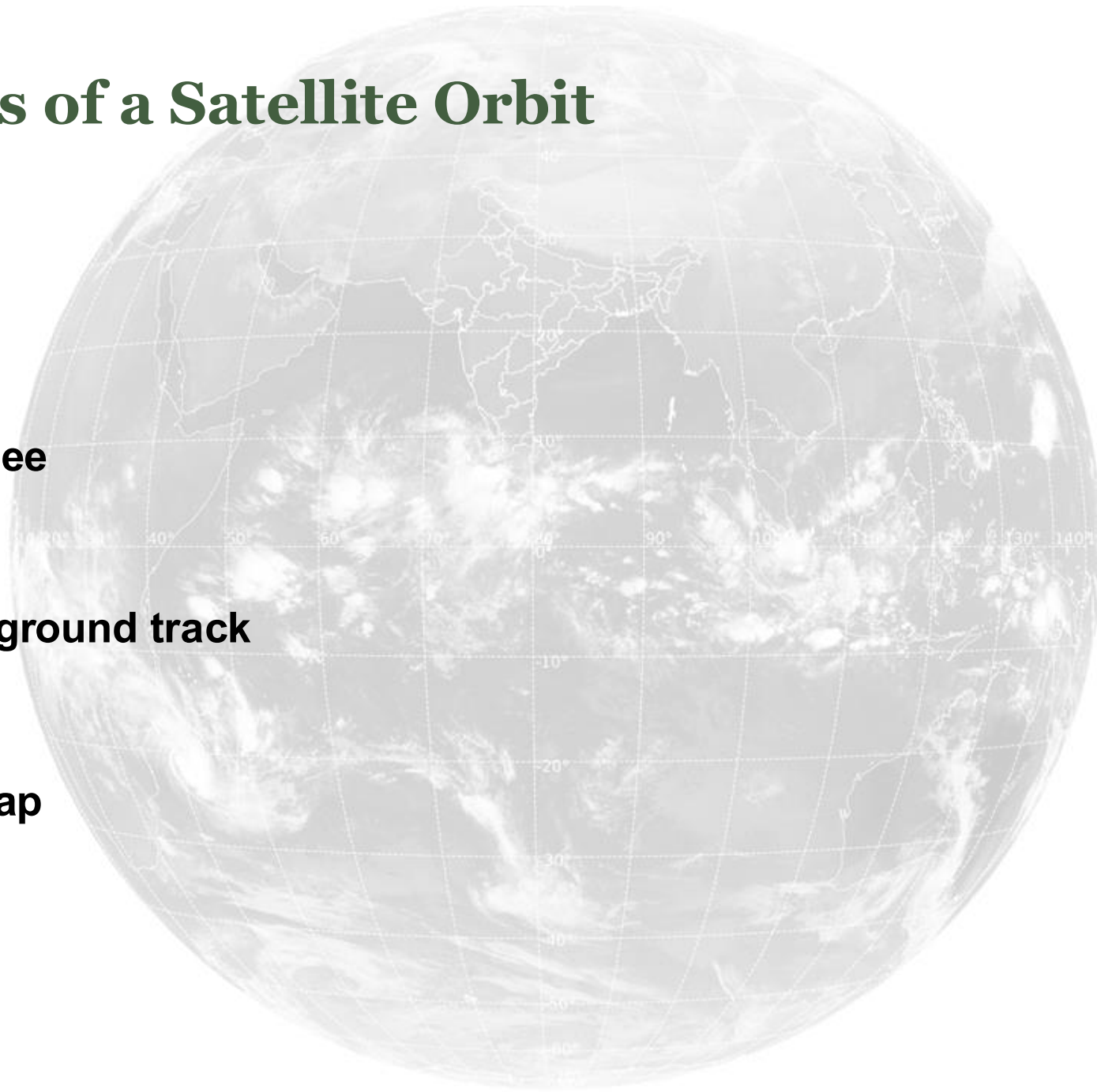
- **Satellite Orbit : The path followed by a satellite in the space**

- Circular or near-circular
- Elliptical



Characteristics of a Satellite Orbit

1. **Orbital period**
2. **Altitude**
3. **Apogee and perigee**
4. **Inclination**
5. **Nadir, zenith and ground track**
6. **Swath**
7. **Sidelap and overlap**



Characteristics of a Satellite Orbit...

1. **Orbital period**

- ❖ **Time taken by a satellite to complete one revolution around the earth**
 - Spatial and temporal coverage of the imagery depends on the orbital period
 - It varies from around 100 minutes to 24 hours

Characteristics of a Satellite Orbit...

2. Altitude

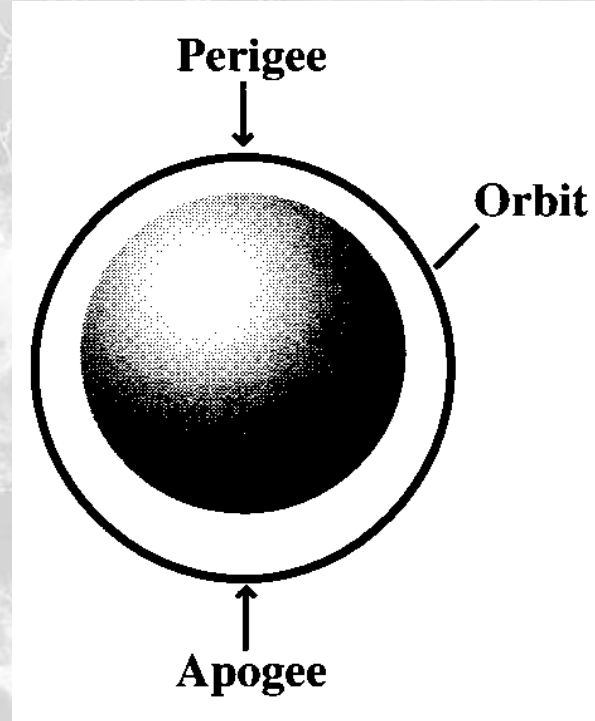
❖ **Altitude of a satellite is its height with respect to the surface immediately below it**

- Low altitude (altitude < 2000 km)
- Moderate altitude
- High altitude (altitude ~36000 km)

Characteristics of a Satellite Orbit...

3. Apogee and perigee

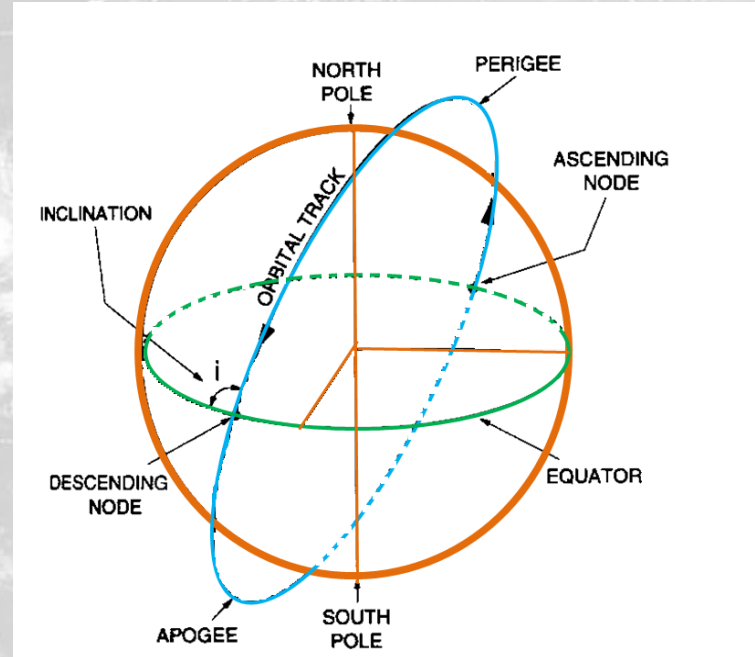
- ❖ Apogee: Point in the orbit where the satellite is at maximum distance from the Earth
- ❖ Perigee: Point in the orbit where the satellite is nearest to the Earth



Characteristics of a Satellite Orbit...

4. Inclination

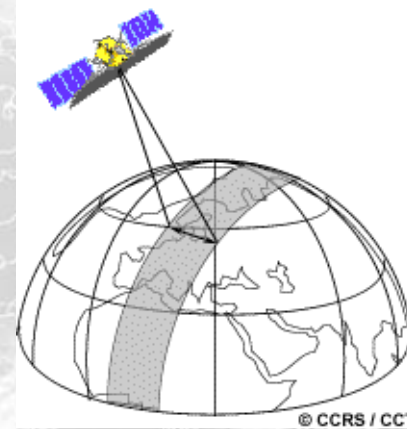
- ❖ Inclination of the orbit is measured clockwise from the equator
- ❖ Inclination of a remote sensing satellite is typically 99°



Characteristics of a Satellite Orbit...

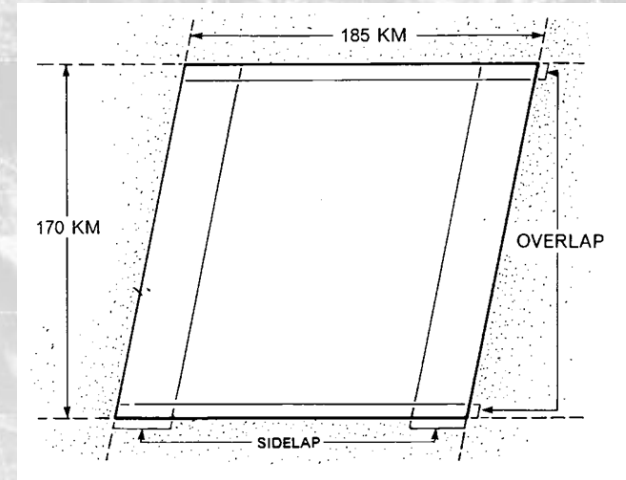
6. Swath

- ❖ Swath of a satellite is the width of the area on the surface of the Earth, which is imaged by the sensor during a single pass



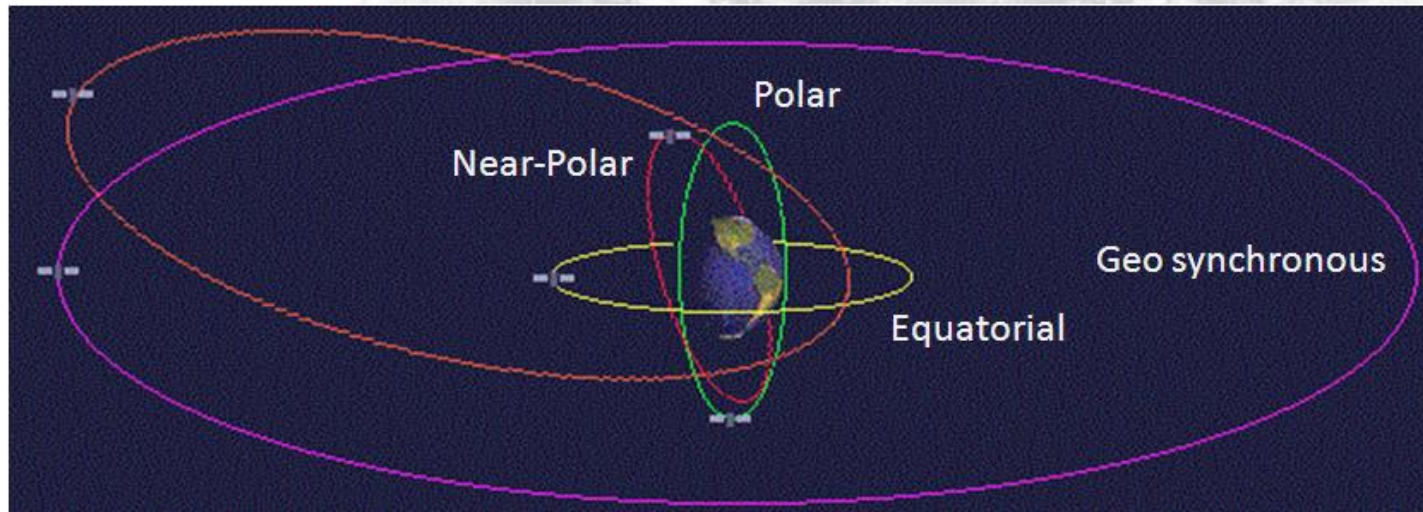
7. Sidelap and Overlap

- ❖ Overlap: Common area on consecutive images along the flight direction
- ❖ Sidelap: Overlapping areas of the images taken in two adjacent flight lines
 - Increase in sidelap helps to achieve more frequent coverage of the areas in the higher latitudes



Types of Satellite Orbits

1. **Geo-synchronous orbit**
2. **Polar orbit**
3. **Sun-synchronous orbit**

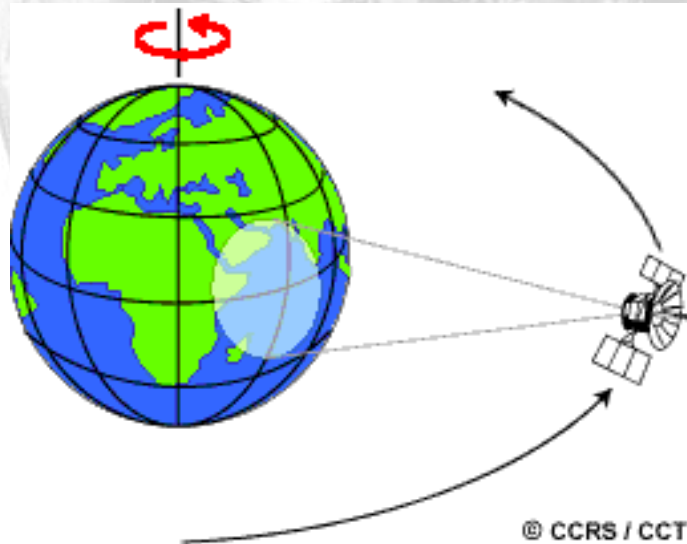


1. Geo-synchronous Orbit

- **Geo-stationary or Geo-synchronous orbit**

❖ From any point on the equator, the satellite appears stationary

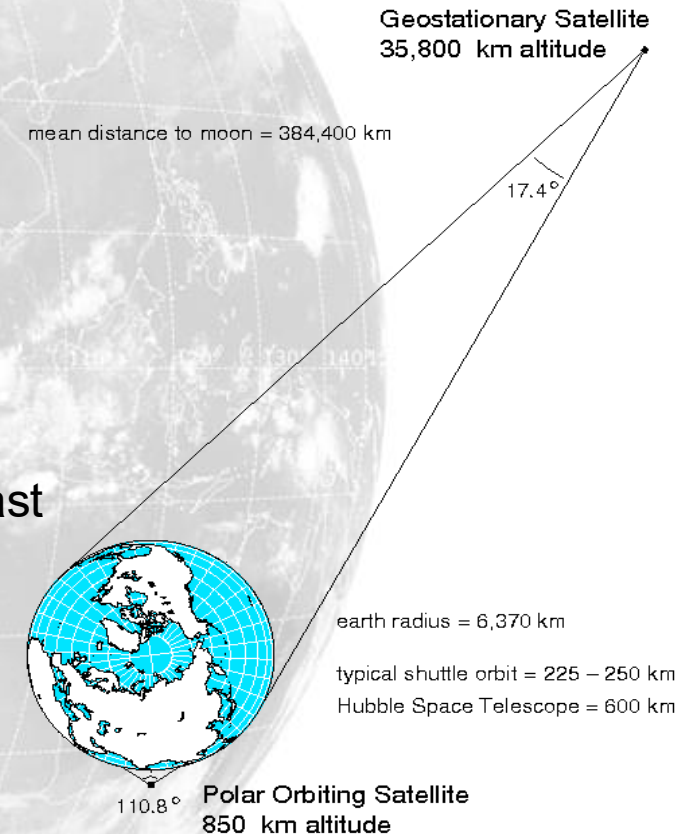
Time required for the satellite
to cover one revolution = Time required for the Earth to
rotate once about its polar axis



Geo-synchronous Orbit...

- **Geo-stationary or Geo-synchronous orbit**

- ❖ Located in the equatorial plane
(Geo-stationary: Inclination = 180°)
(Geo-synchronous: Inclination need not be = 180°)
- ❖ Located above any particular longitude to view any selected region
- ❖ Satellite revolves in the same direction as that of the Earth : West to East
- ❖ Geo-synchronous orbits are at very high altitude : $\sim 36,000$ km
- ❖ Foot prints are very high : 45-50% of the globe



Geo-synchronous Orbit...

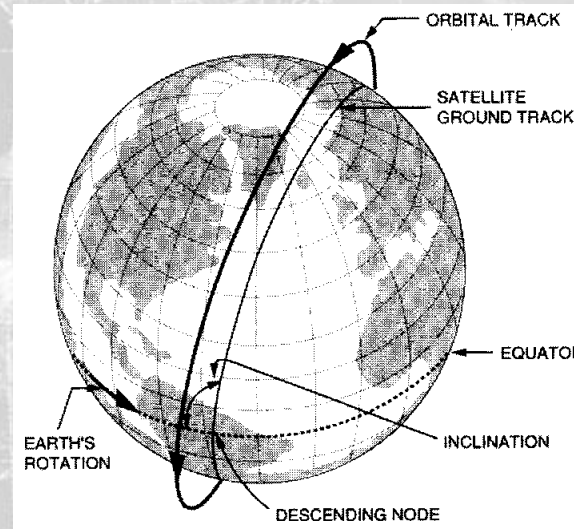
- **Geo-stationary or Geo-synchronous satellites**
 - ❖ Communication and meteorological applications
 - GOES
 - GOES East at 75° W and GOES West at 135° W longitude
 - GMS
 - Located over New Guinea
 - MeteoSAT
 - Located over Europe
 - INSAT
 - INSAT-3A located at 93.5° E longitude

2.Polar Orbit

- **Polar or Near-Polar orbit**

- ❖ **Inclined at nearly 90 degrees**

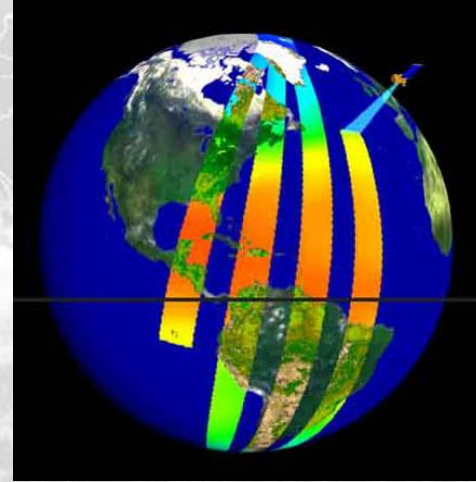
- Usually low altitude orbits
 - Typically 700-800 km
 - Orbit period is less
 - Typically 90-103 min
 - Satellites make more than one revolution around the Earth in a single day



Polar Orbit...

- **Polar or Near-Polar orbit**

- ❖ Makes more than one revolution in a single day
- ❖ Due to the Earth's rotation, in each revolution the satellite passes over different areas
- ❖ Gives complete coverage of the earth's surface during an orbit cycle
 - **Orbit cycle:** When the nadir point of the satellite passes over the same point on the Earth's surface for a second time
 - Orbit cycle is also known as repeat cycle



Polar Orbit...

- **Revisit period:** Time lapsed between two successive views of the same area by a satellite
 - Revisit time need not be the same as the orbit cycle
 - Off-nadir viewing capabilities using steerable sensors helps to achieve revisit time less than the orbit cycle
 - In near-polar orbits, orbit paths come closer together near the poles.
 - More overlap in adjacent swaths at high latitude
 - More frequent imaging near the polar region compared to the equatorial zone

3. Sun-synchronous Orbit

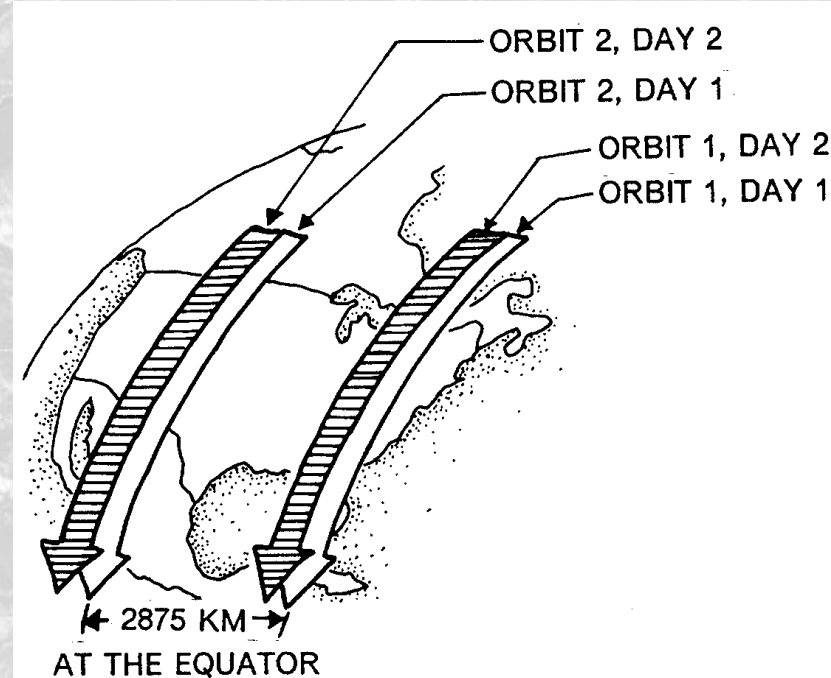
- **Sun-synchronous orbits**

- Special case of polar orbit
- Satellite passes over the same part of the Earth at roughly at the same local time each day
- Used for satellites that need a constant amount of sunlight
- Satellites revolving in sun-synchronous orbits : Landsat, IRS satellites

Sun-synchronous Orbit ...

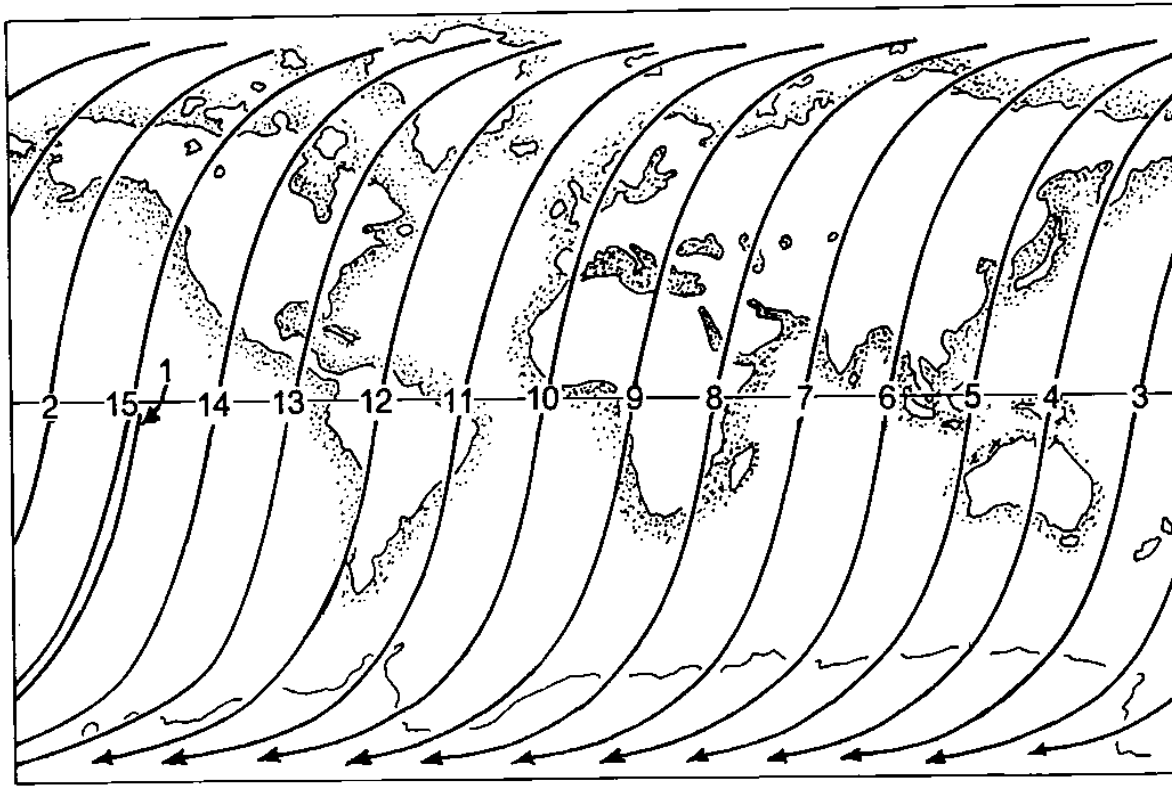
- **Typical Sun-synchronous satellite**

- Completes 14 orbits a day
- Successive orbits are shifted by around 2875 km at the equator
- Satellite's path is shifted in longitude by 1.17° everyday towards west



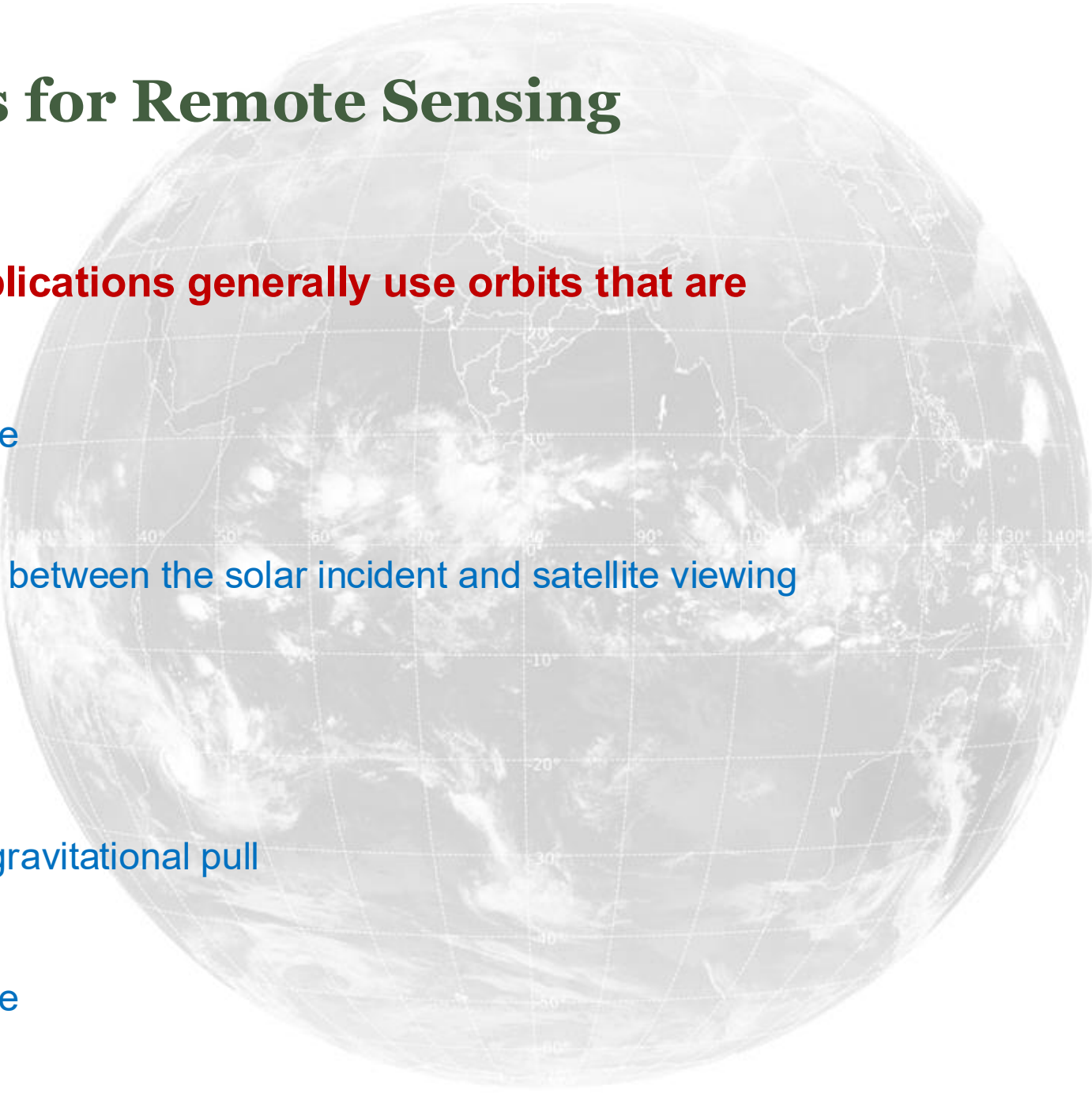
Sun-synchronous Orbit ...

- Landsat orbits

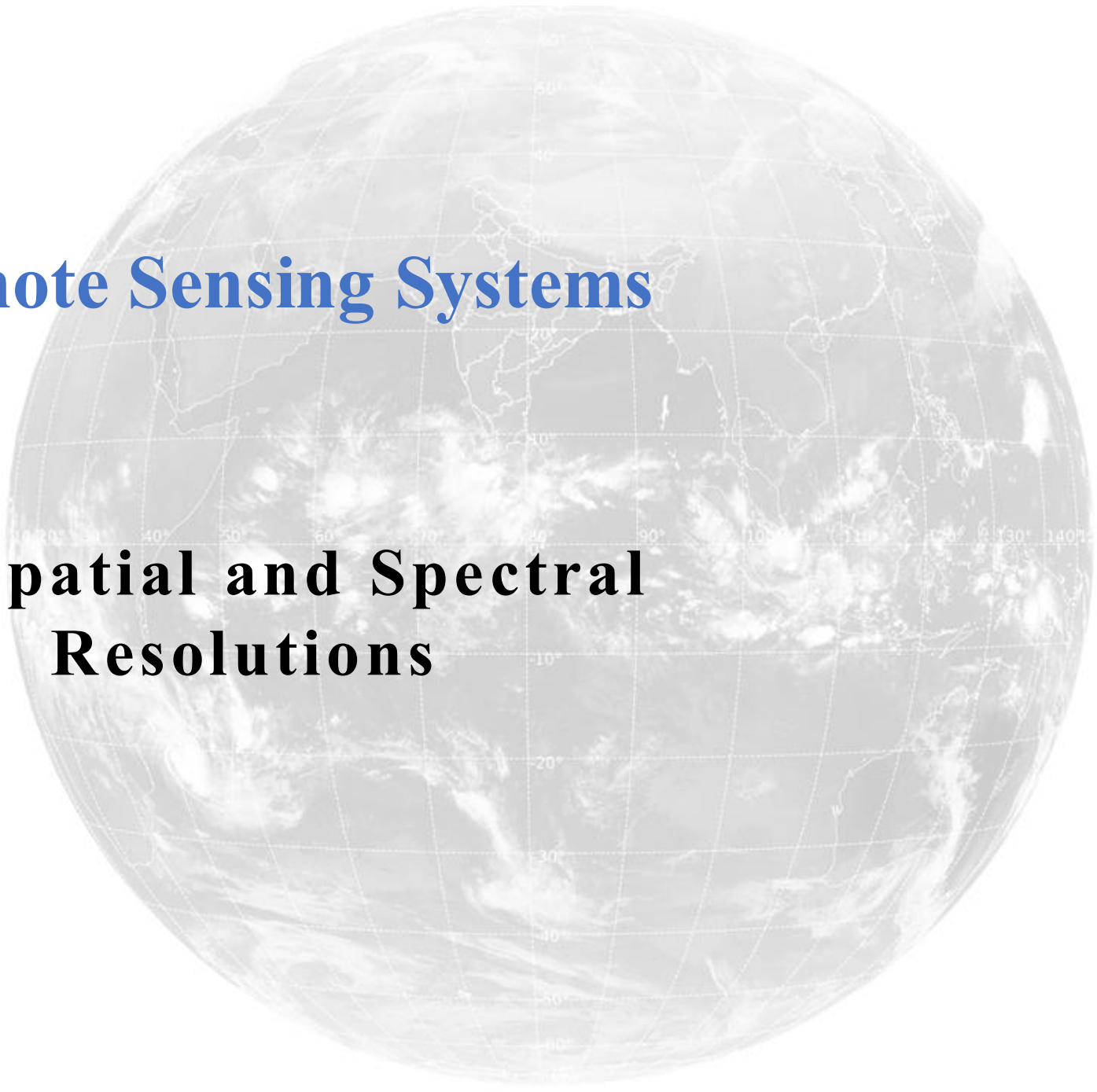


14 orbits per day, Repeat cycle = 18 days

Satellite Orbits for Remote Sensing



- **Remote sensing applications generally use orbits that are**
 - **Near polar**
 - For global coverage
 - **Sun-synchronous**
 - For constant angle between the solar incident and satellite viewing
 - **Near circular**
 - For uniform swath
 - **Inclined**
 - For differences in gravitational pull
 - **Medium period**
 - For global coverage



Remote Sensing Systems

(ii) Spatial and Spectral Resolutions

Introduction



- In remote sensing **resolution** means the **resolving power**
 - Capability to identify the presence of two objects
 - Capability to identify the properties of the two objects
- An image that shows finer details is said to be of finer resolution compared to the image that shows coarser details

Types of Resolution

- **4 types** of resolutions are defined for the remote sensing systems
 - ❖ Spatial resolution
 - ❖ Spectral resolution
 - ❖ Temporal resolution
 - ❖ Radiometric resolution

Spatial Resolution

- Spatial resolution: Size of the smallest dimension on the Earth's surface over which an independent measurement can be made by the sensor
 - Expressed by the size of the pixel on the ground in meters
 - Controlled by the Instantaneous Field of View (IFOV)

Coarse Spatial Resolution



Fine Spatial Resolution

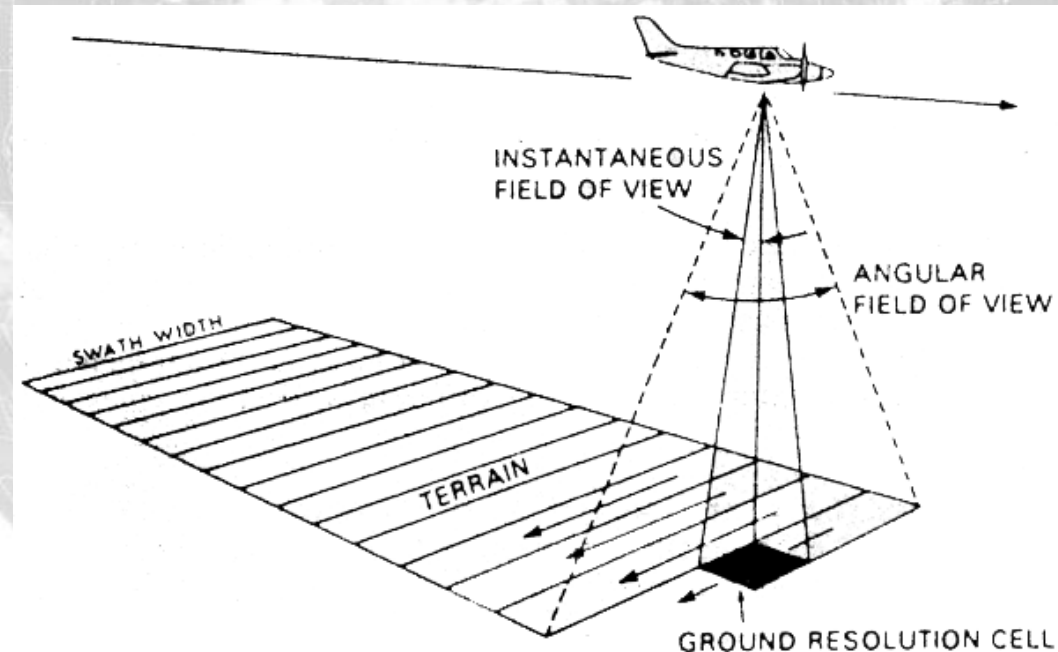


Instantaneous Field of View

- **IFOV: Instantaneous Field of View**

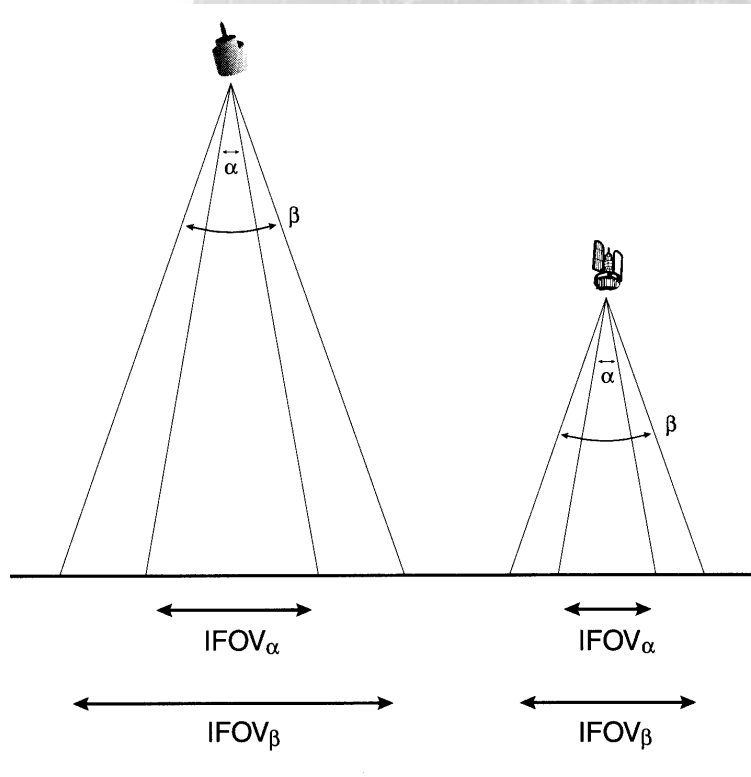
- ❖ Angular cone of visibility of the sensor
- ❖ Area on the Earth's surface that is seen at one particular moment of time

- ❖ IFOV depends on
 - Altitude of the sensor above the ground level
 - Viewing angle of the sensor



Instantaneous Field of View...

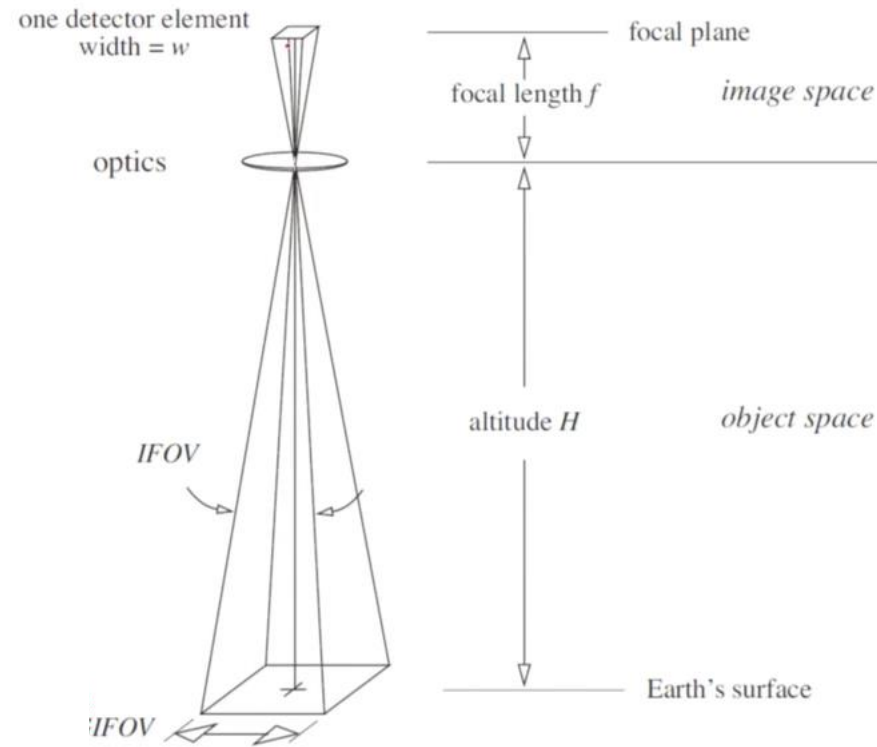
- A narrow viewing angle produces a smaller IFOV
- IFOV increases with altitude of the sensor



Ground Resolution Cell

- Ground Resolution or Ground Resolution Cell : Size of the area viewed by the sensor on the ground at one particular moment of time
- Depends on
 - Altitude of the sensor
 - IFOV of the sensor
- Obtained by multiplying the IFOV (in radians) by the distance from the ground to the sensor.
- It is also referred as the spatial resolution of the remote sensing system

Simple geometry



- IFOV
- GIFOV

$$IFOV = 2 \operatorname{atan}\left(\frac{w}{2f}\right)$$

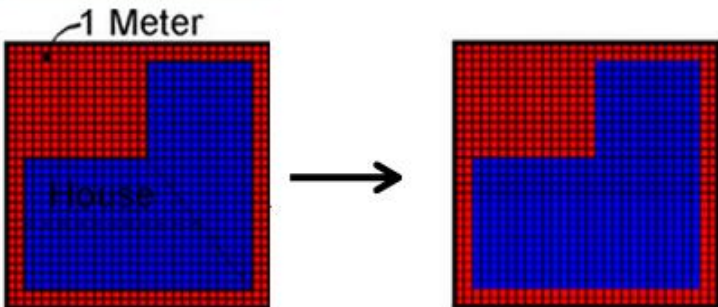
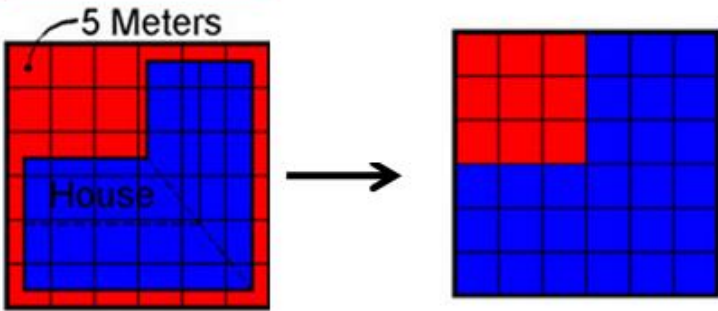
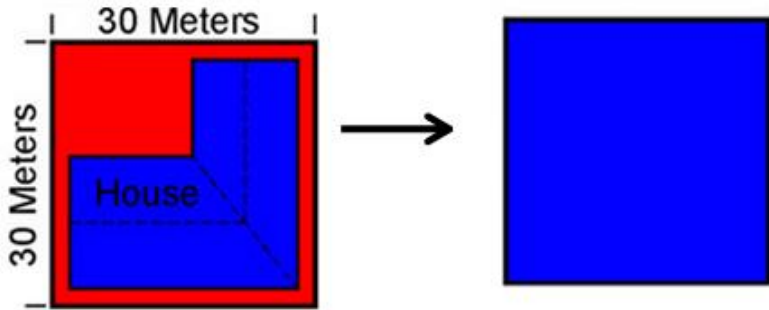
$$GIFOV = 2H \tan\left(\frac{IFOV}{2}\right)$$

Spatial Resolution and Feature Identification

- For a feature to be detected, its size generally has to be equal to or larger than the resolution cell
- If more than one feature is present within ground resolution cell, the signal response is a mixture of the signals from all the features
 - From the average brightness recorded, any one particular feature among them may not be detectable
 - Smaller features may sometimes be detectable if their reflectance dominates within a particular resolution cell

Spatial Resolution and Feature Identification...

Pixel size (Resolution) Pixel output (Display)



Example

Signature from the “house” dominates for the cell and hence the entire cell is classified as “house”

Shape and spatial extent of the feature is better captured. However, some discrepancy is present along the boundary

Feature shape and the spatial extent is more precisely identified

Classes of Spatial Resolution

- **Low resolution systems**

- Spatial resolution $> 1\text{km}$
- MODIS, AVHRR

- **Medium resolution systems**

- Spatial resolution is $100\text{m} - 1\text{km}$
- IRS WiFS (188m), Landsat TM-Band 6 (120m), MODIS-Bands 1-7 (250-500m)

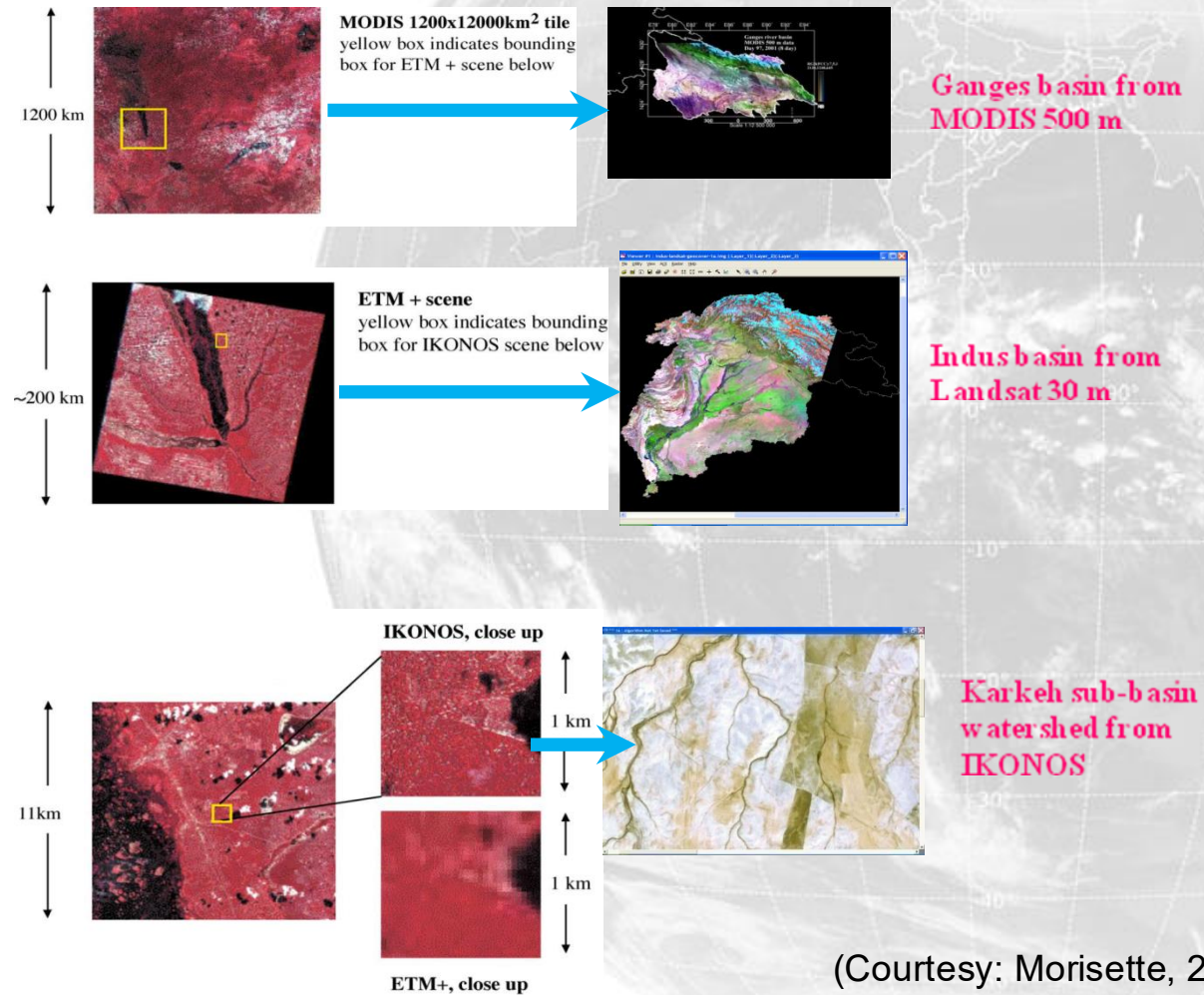
- **High resolution systems**

- Spatial resolution approximately in the range $5\text{-}100\text{m}$
- Landsat ETM+ (30m), IRS LISS-III (23m MSS, 6m Panchromatic), IRS AWiFS (56-70m), SPOT 5(2.5-5m Panchromatic)

- **Very high resolution systems**

- Spatial resolution less than 5m
- GeoEye (0.45m for Panchromatic, 1.65m for MSS), IKONOS (0.8-1m Panchromatic), Quickbird (2.4-2.8 m)

Spatial Resolutions and Scale of Applicability

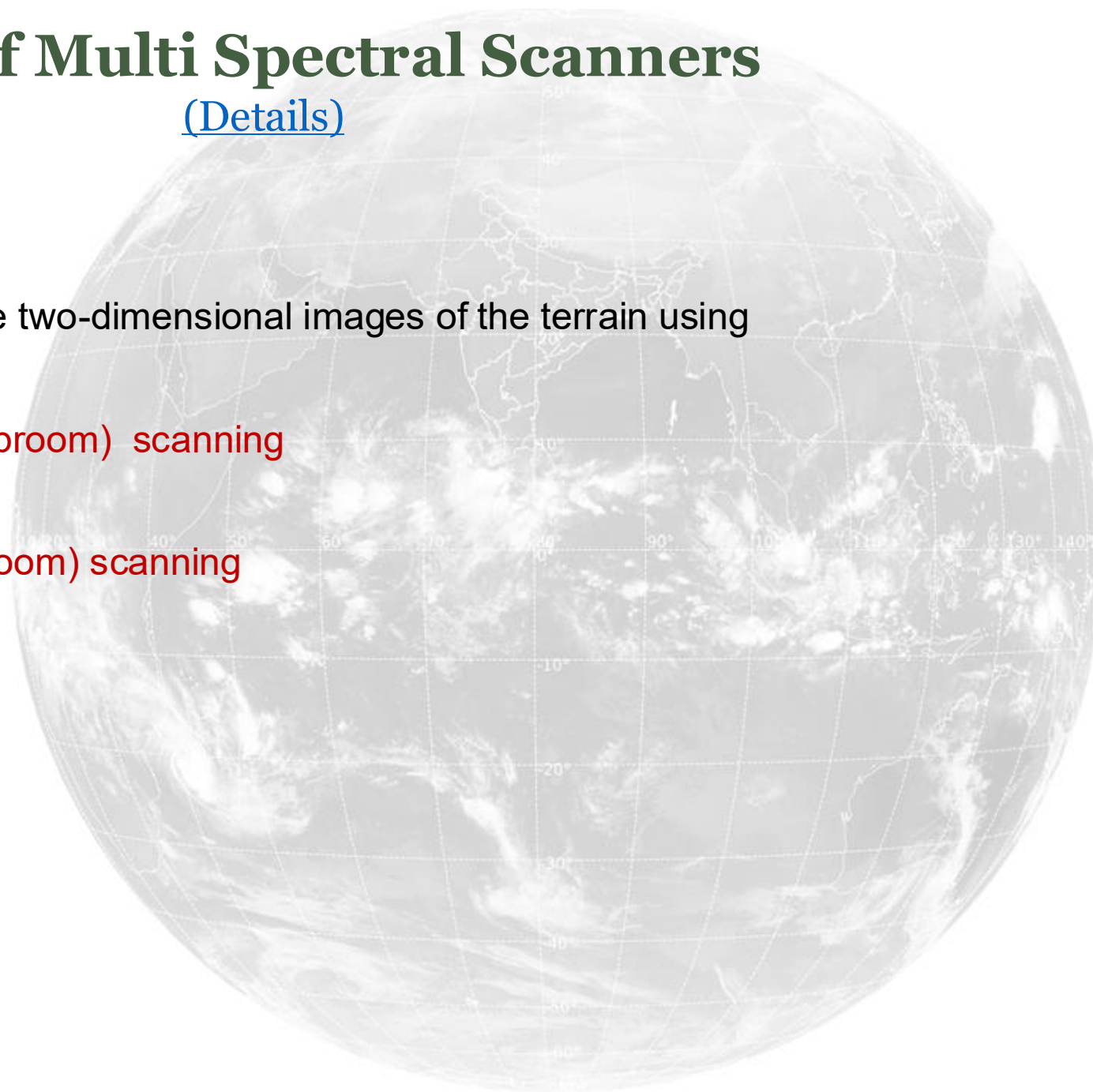


(Courtesy: Morisette, 2002)

Types of Multi Spectral Scanners

(Details)

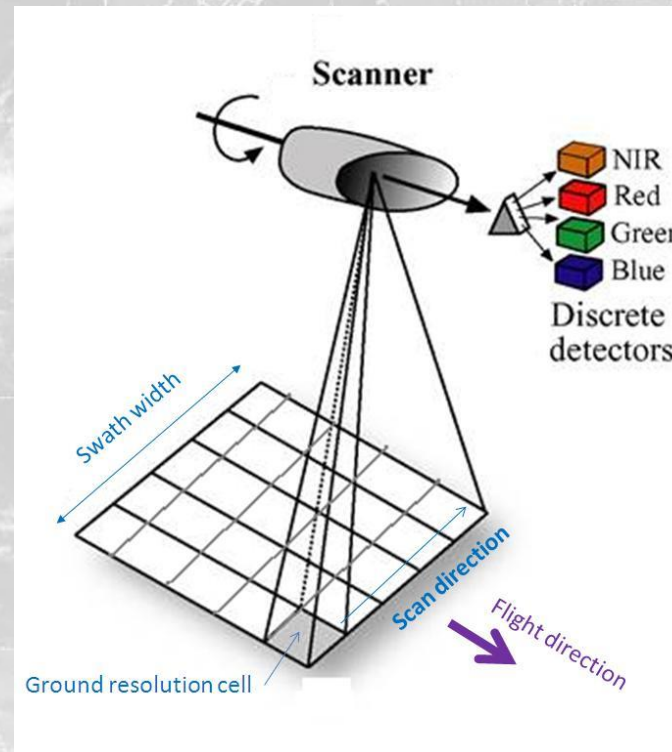
- MSS systems generate two-dimensional images of the terrain using
 - ❖ Across-track (whiskbroom) scanning
 - ❖ Along-track (pushbroom) scanning



Across-Track Scanning

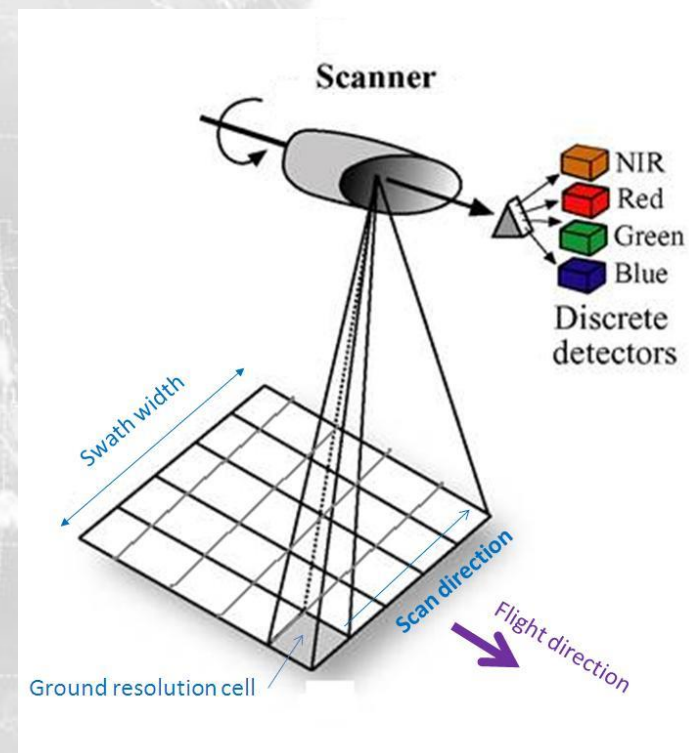
- Across-track scanner is also known as whisk-broom scanner

- Rotating or oscillating mirrors are used
- Scan lines at right angles to the flight line are scanned successively as the platform moves forward
- Continuously measures the energy from one side to the other side of the platform



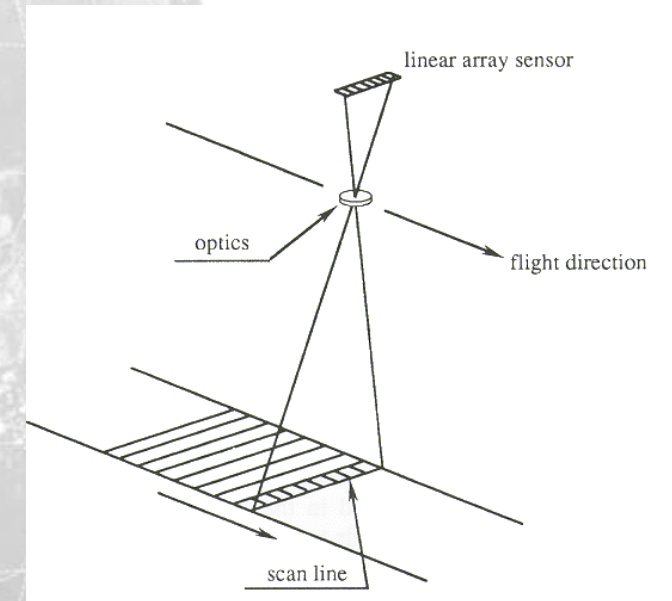
Across-track scanning...

- Incoming radiation is separated into several thermal and non-thermal wavelength components using **a dichroic grating and a prism**
- **An array of electro-optical detectors**, each having peak spectral sensitivity in a specific wavelength band, is used to measure each wavelength band separately



Along-track scanning

- Along-track scanner is also known as push-broom scanner.
- No scanning mirrors are used
- A linear array of detectors is used to simultaneously record the energy received from multiple ground resolution cells along the scan line
- The array of detectors are pushed along the flight direction to scan the successive scan lines → push-broom scanner
- Size of the ground resolution cell is determined by the IFOV of a single detector

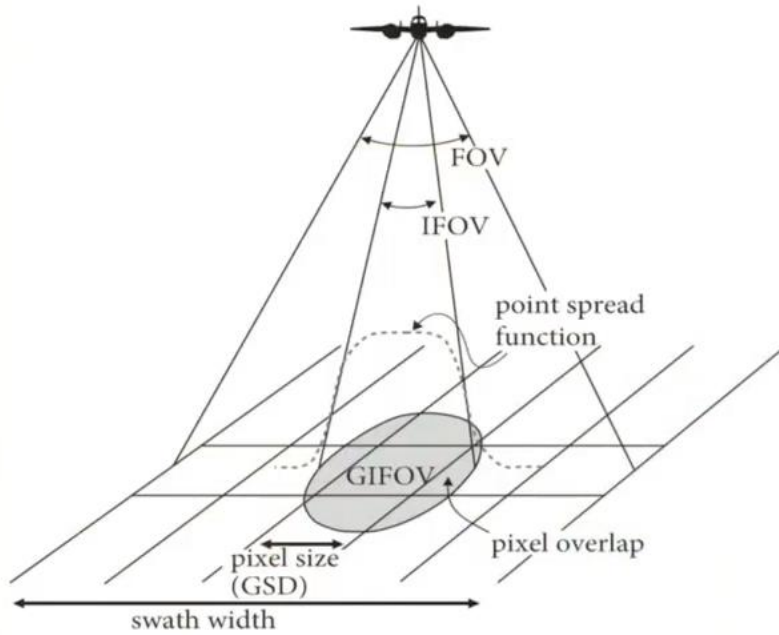


(Source: <http://stlab.iis.u-tokyo.ac.jp/>)

Along-track scanning... [Details](#)

- The linear array of detectors consists of numerous (more than 10,000) charged coupled devices (CCDs)
- Each linear array is dedicated to record energy in a single band
- Each detector element is dedicated to record the energy in a single column
- The arrays of detectors are arranged in the focal plane of the scanner such that each scan line is viewed simultaneously by all the arrays
- Advantages of along-track scanning
 - Linear array of detectors provides longer dwell time over each ground resolution cell
 - Higher signal strength
 - Finer radiometric resolution
 - Finer spatial and spectral resolution without impacting radiometric resolution

Simple geometry



- FOV
- GFOV (Swath width)

$$GFOV = 2H \tan\left(\frac{FOV}{2}\right)$$

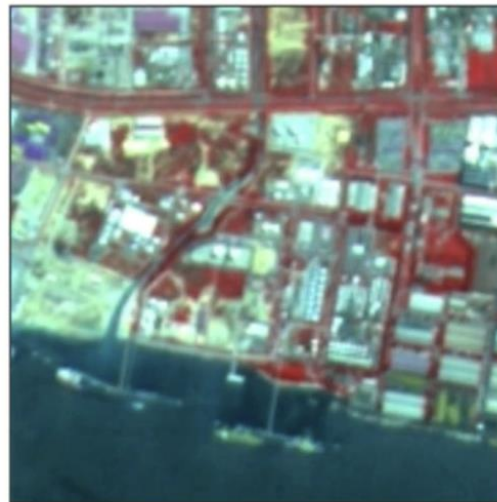
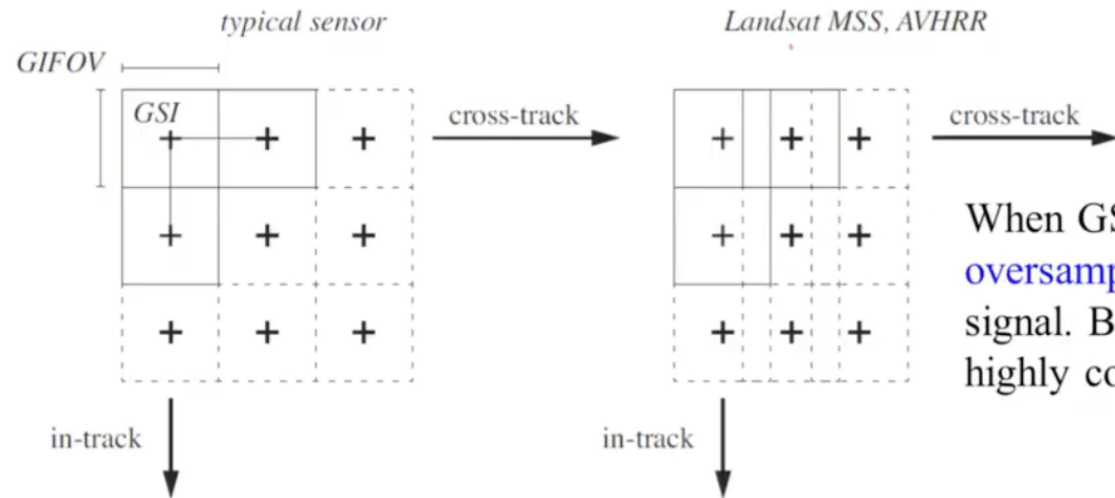
Spatial resolution and Pixel Size

$$GSI = \text{inter-detector spacing} \times \frac{H}{f}$$

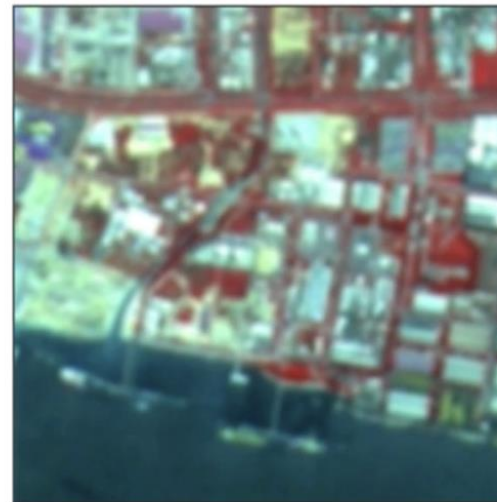
- Image spatial resolution is related to the GIFOV, as the sensor averages all the energy available within that GIFOV.
- However, the pixel size is fixed by the Ground projected sample interval (GSI) or ground projected sample distance (GSD).
- GSI is related to the spacing between detectors or the [rate at which we sample the signal from the detector](#).
- A [pixel](#) is defined as the data sample in the output product to which a radiance is assigned.
- A pixel can have dimensions different from that of the system's footprint (GIFOV).

How GSI and GIFOV are related?

Generally they are kept equal but not always.



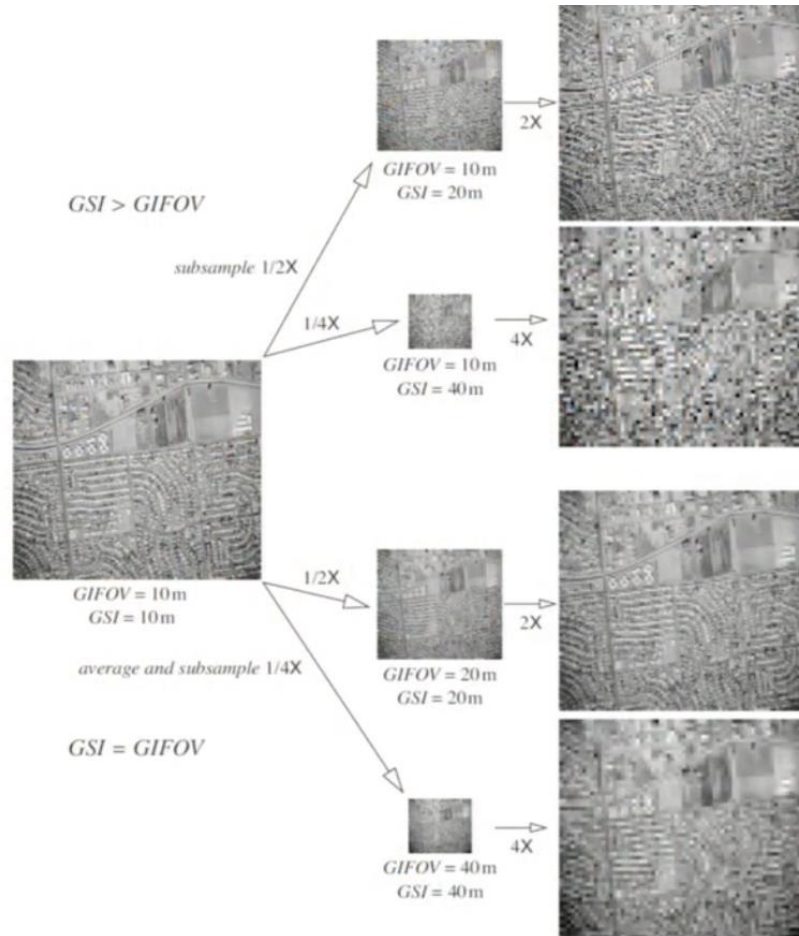
10 m resolution, 10 m pixel size



30 m resolution, 10 m pixel size



80 m resolution, 10 m pixel size



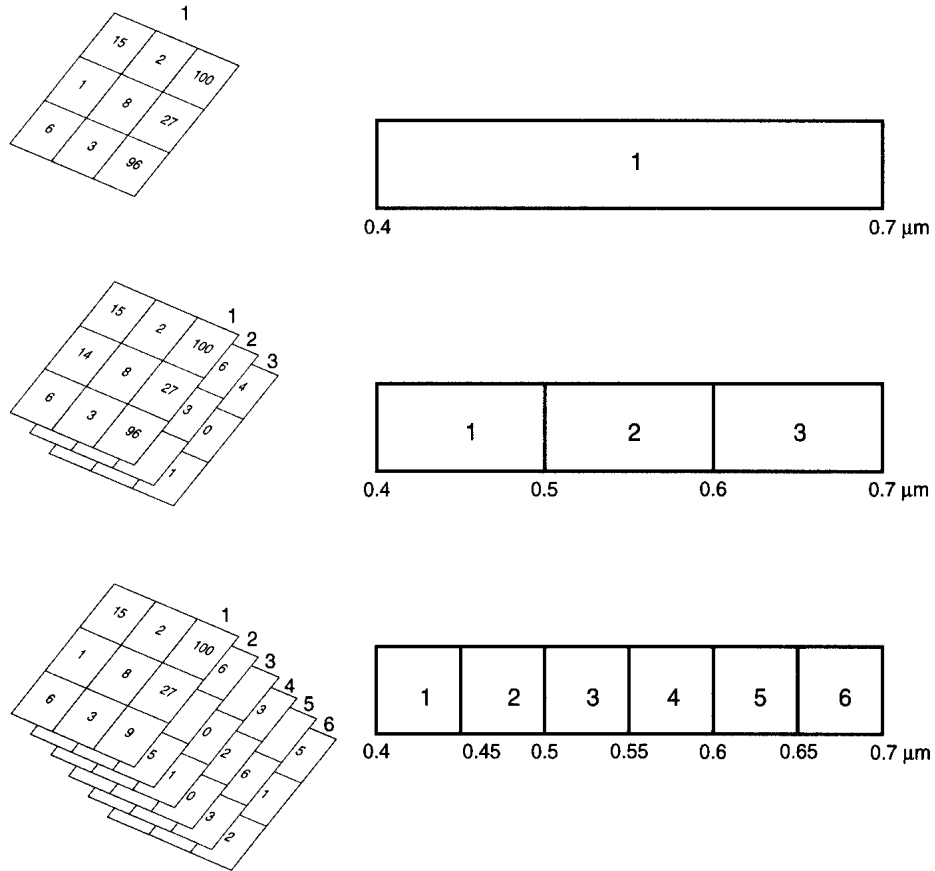
When $GSI > GIFOV$, it causes under sampling and will lead to **spatial aliasing**

Spectral Resolution

- **Spectral resolution**
 - ❖ Ability of a sensor to define fine wavelength intervals
 - ❖ Ability of a sensor to resolve the energy received in a spectral bandwidth to characterize different constituents of earth surface
- Depends on
 - Spectral band width of the filter
 - Sensitiveness of the detector
- The finer the spectral resolution, the narrower the wavelength range for a particular channel or band

Spectral Resolution...

Finer the spectral resolution, the narrower the wavelength range for a particular channel or band

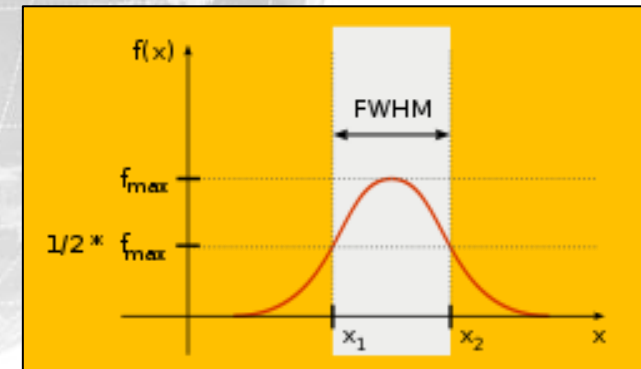
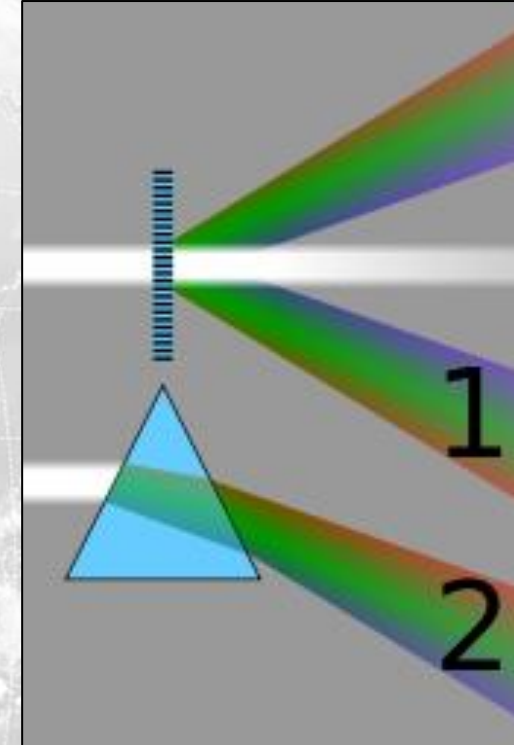


SPECTRAL SENSITIVITY

- ❑ An optical sensor converts light into electronic signals
- ❑ Optical sensors use **prisms / filters** to separate the light into spectral regions
- ❑ Filters are specialized glass materials that pass certain wavelengths and absorb / block all other wavelengths
- ❑ Filters are manufactured by adding DYES to glass
- ❑ In Visible region, filters are generally used to separate out BLUE (due to shorter wavelength – higher scattering)
- ❑ A deep red filter blocks all visible radiation, but allows to pass infra red radiation (for vegetation growth)
- ❑ More than filters, DIFFRACTION GRATINGS are used in satellite sensors due to their effectiveness, small size, light weight

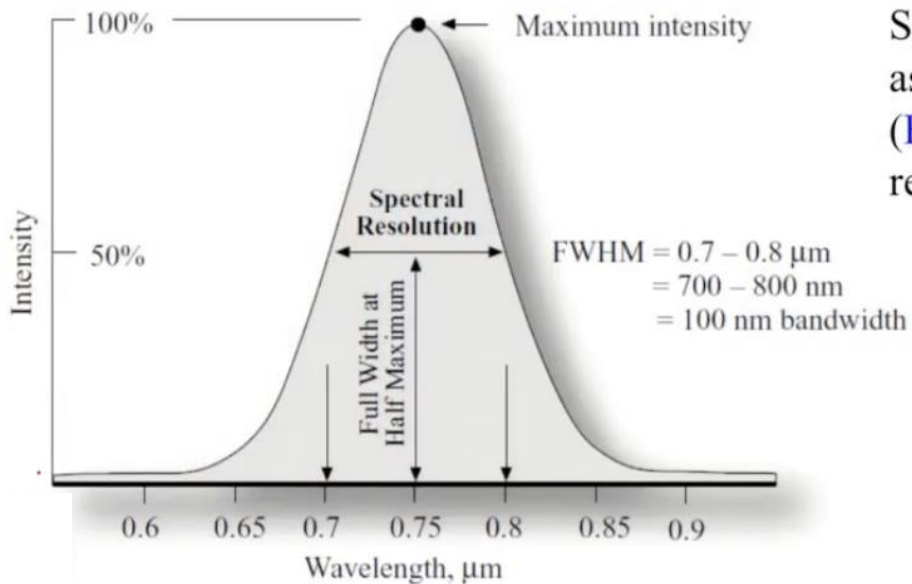
SPECTRAL SENSITIVITY

- ❑ Diffraction grating is an optical component with a periodic structure, which splits and diffracts light (EM wave) into several beams travelling in different directions
- ❑ Due to the use of filters / diffracting gratings used by sensors to define spectral limits, spectral sensitivity varies across a specified defined interval
- ❑ Spectral sensitivity is the relative efficiency of detection of light as a function of wavelength
- ❑ Spectral sensitivity of a sensor is generally specified using “Full Width Half Maximum
- ❑ Beyond the limits of FWHM, the response (measurement) is so weak

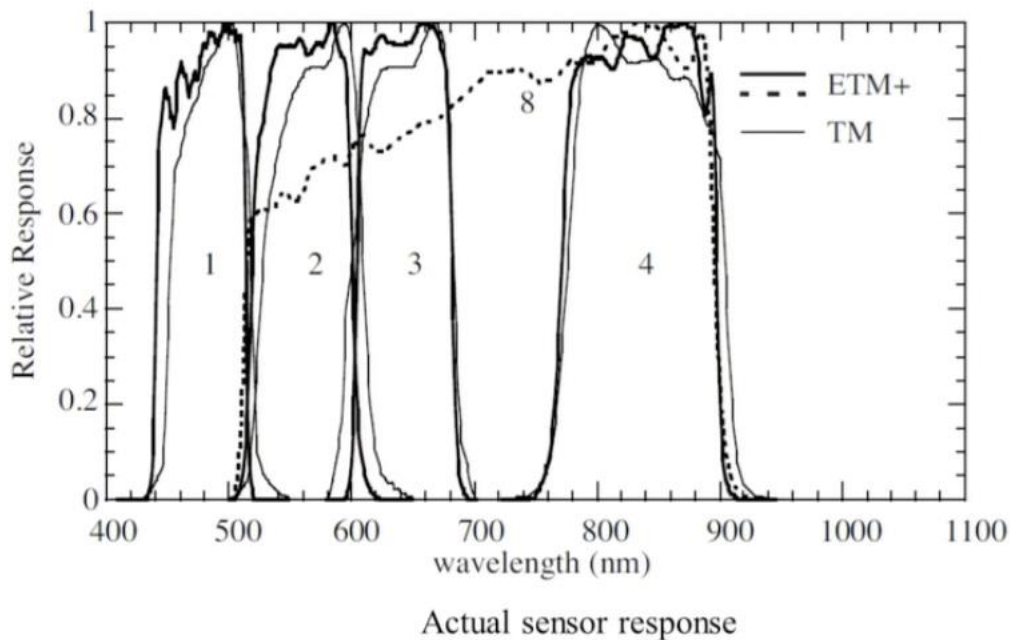


Spectral bandwidth

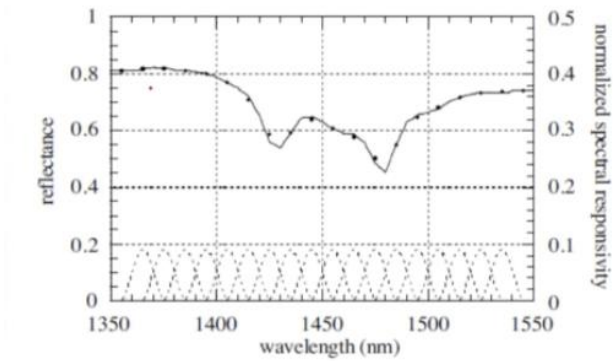
Spectral Resolution



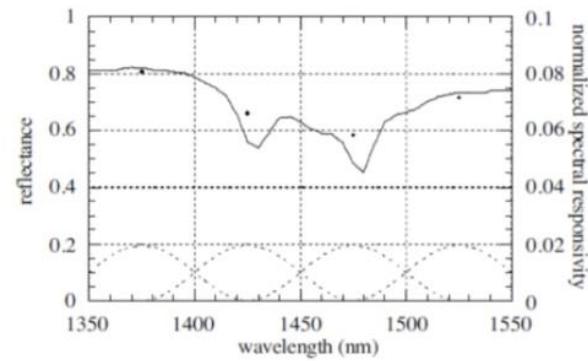
Spectral bandwidth is often calculated as full width at half maximum (FWHM) assuming a gaussian spectral response curve.



Spectral resolution



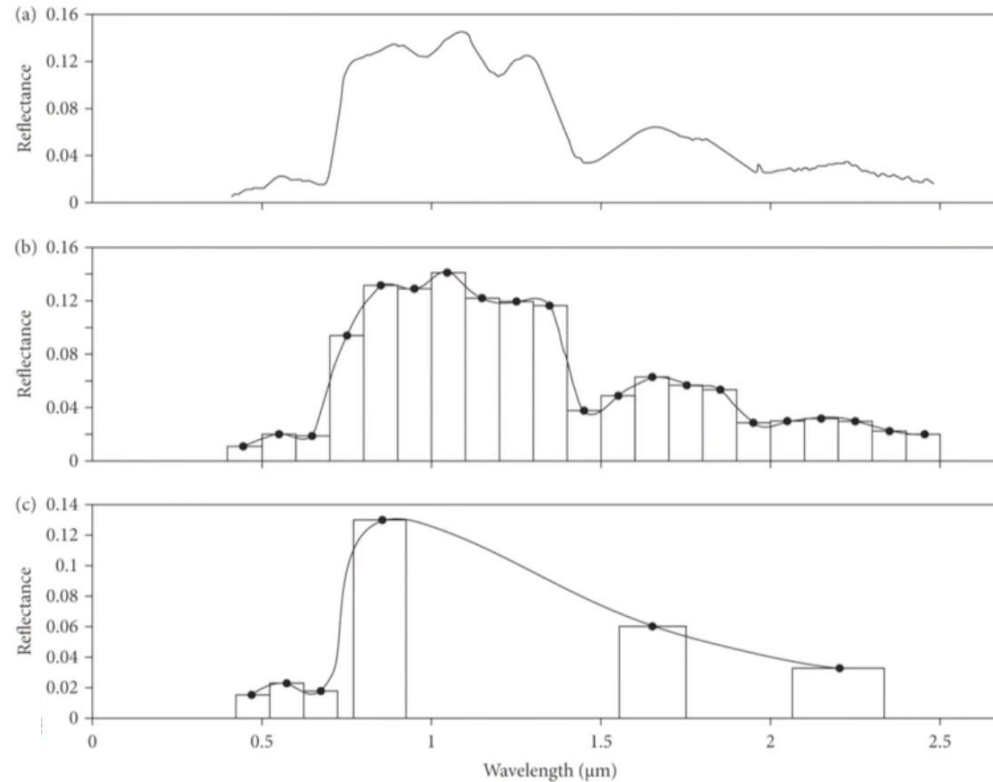
10nm spectral bandwidth



50nm spectral bandwidth

The location of central wavelength and the spectral bandwidth affects the radiance we collect from the earth's surface.

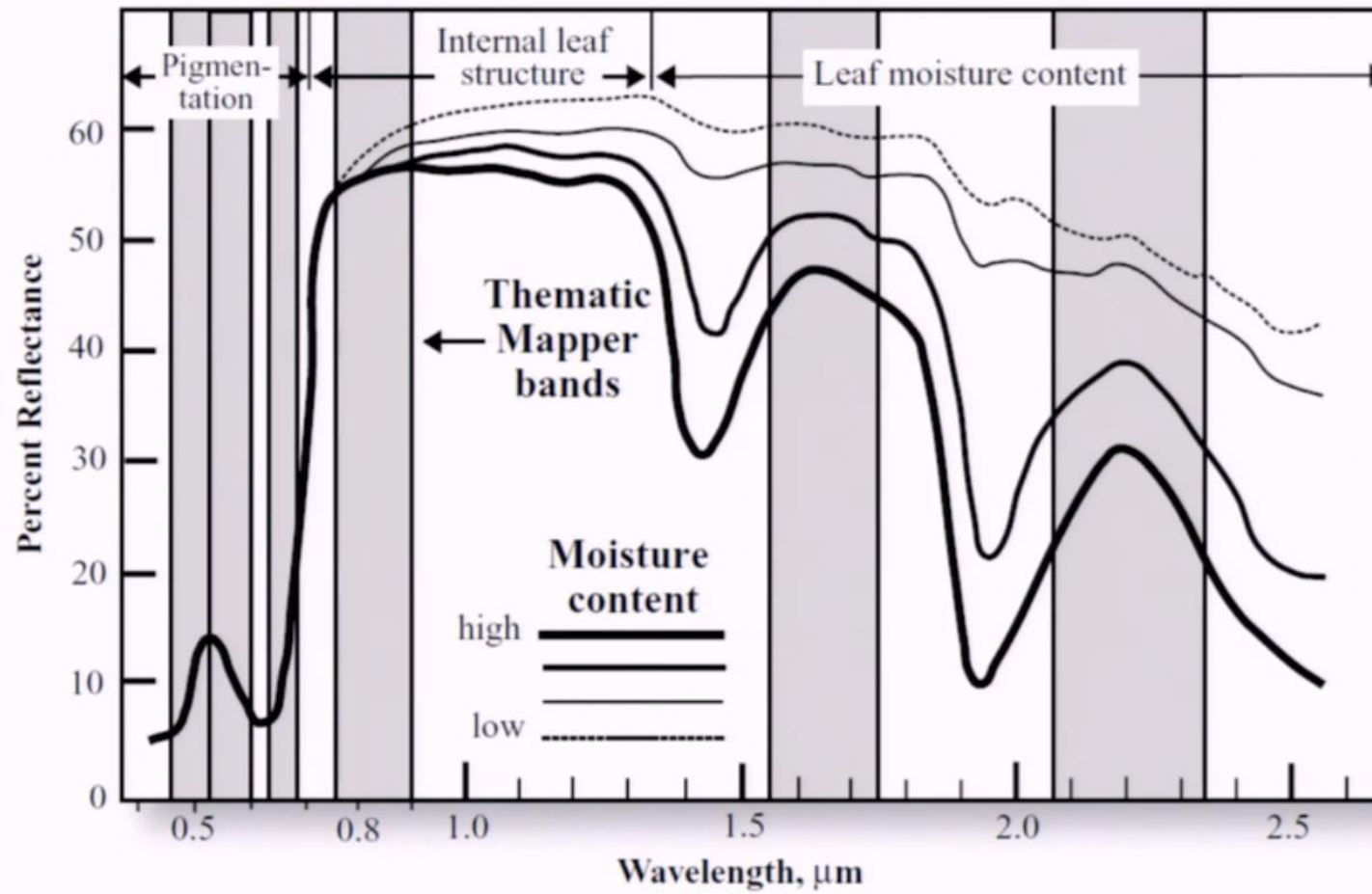
Spectral resolution



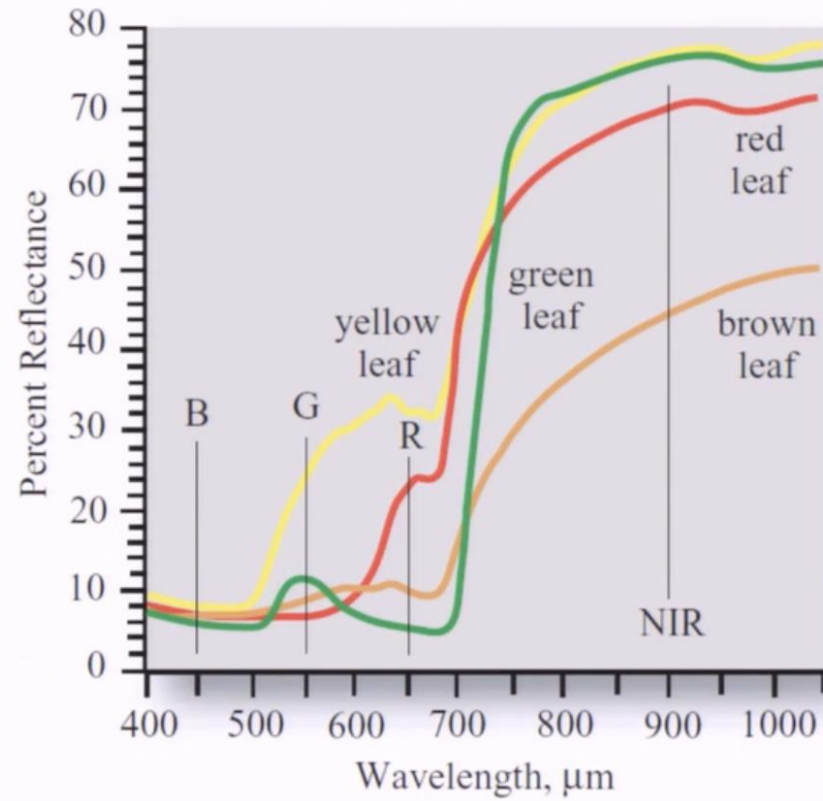
The location, bandwidth and the number of bands affects our ability in identifying earth surface features.

Observing vegetation from Landsat TM

Reflectance of a Sycamore Leaf at Different Moisture Conditions



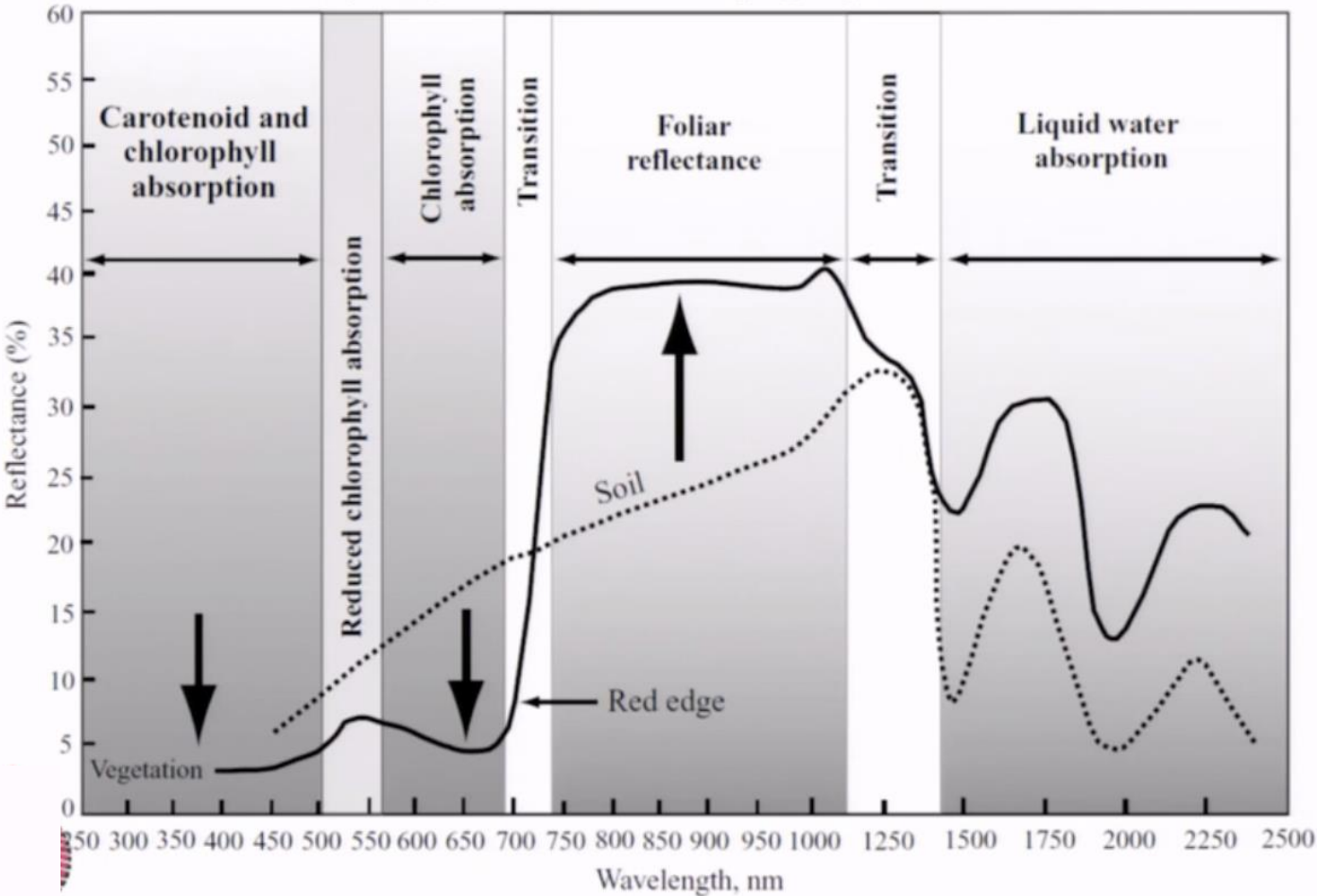
Reflectance curves for leaves in varying senescence stages



1:

Red Edge

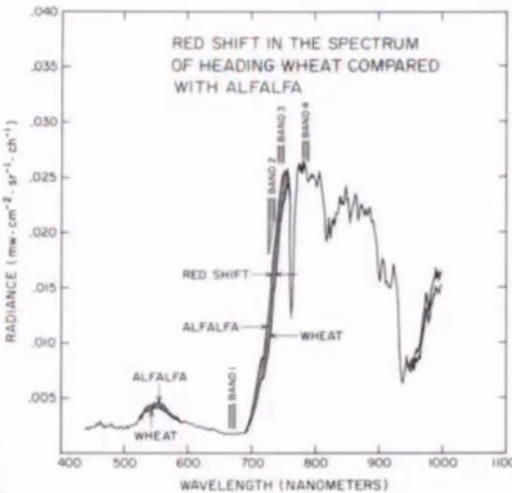
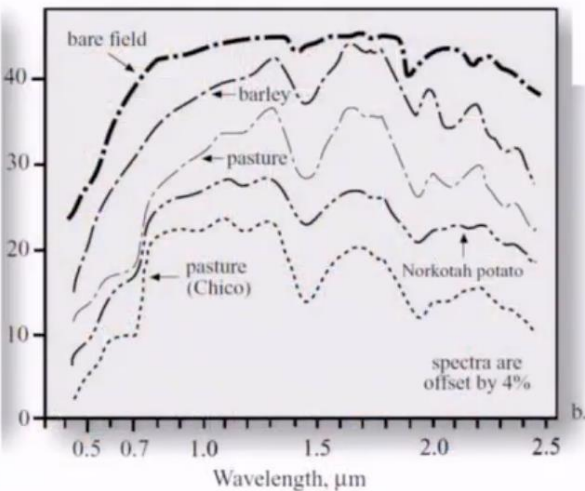
Physiological Basis for Developing Vegetation Indices



Red Edge

The red edge may undergo:

- Red shift
- Blue shift

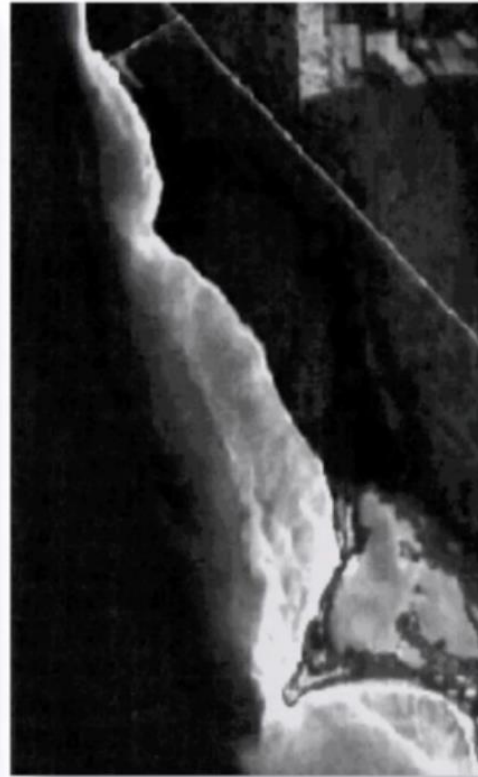


(Collins, 1978)

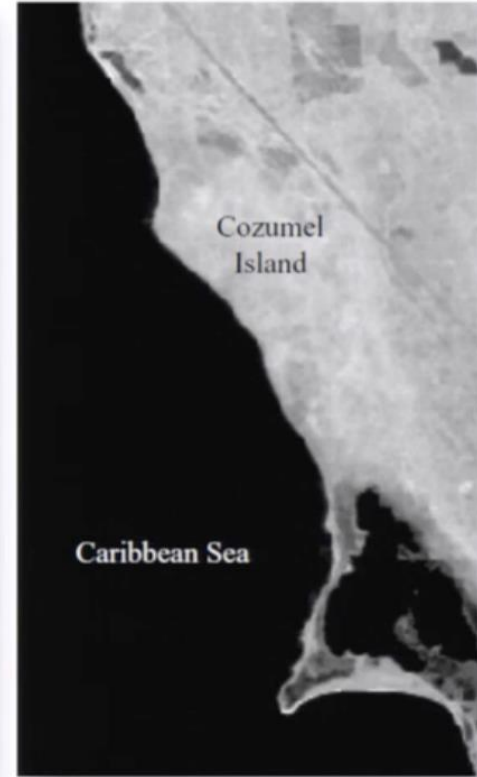
Example



a. SPOT Band 1 (0.50 - 0.59 μm).



b. SPOT Band 2 (0.61 - 0.68 μm).



c. SPOT Band 3 (0.791 - 0.89 μm).

SPECTRAL Resolution



1) Panchromatic Sensing system

- Sensor is a single channel detector sensitive to radiation within a broad wavelength range
- If the wavelength range coincide with the visible range, then the resulting image resembles a "black-and-white" photograph taken from space
- The spectral information or "colour" of the targets is lost

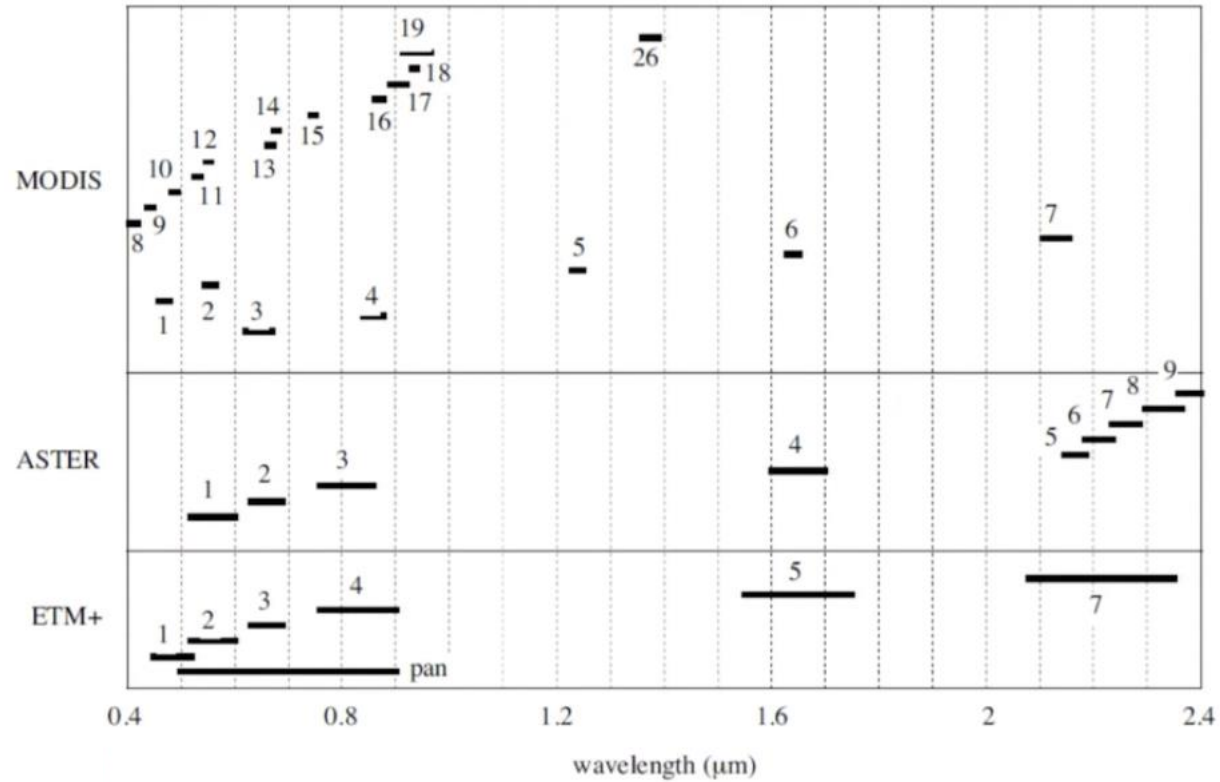
2) Multispectral Sensing system

- Sensor is a multichannel detector with a few spectral bands
- Each channel is sensitive to radiation within a narrow wavelength band
- The resulting image is a multilayer image which contains both the brightness and spectral (colour) information of the targets being observed

3) Hyperspectral Sensing system

- Acquires images in about a hundred or more **contiguous spectral** bands
- Enables better characterization and identification of targets

Spectral characteristics of some sensors

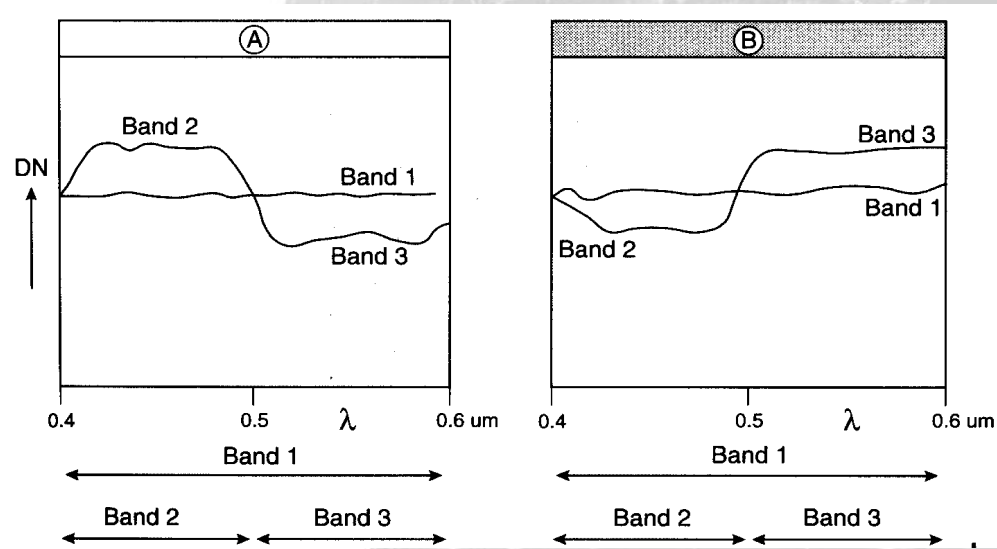


Spectral Resolution...

- ❖ Most of the remote sensing systems are multi-spectral, using more than one spectral band
- ❖ Spectral resolution of some of the remote sensing systems
 - IRS LISS-III uses 4 bands: 0.52-0.59 (green), 0.62-0.68 (red), 0.77-0.86 (near IR) and 1.55-1.70 (mid-IR).
 - The Aqua/Terra MODIS instruments use 36 spectral bands, including three in the visible spectrum.
 - Recent development is the hyper-spectral sensors, which detect hundreds of very narrow spectral bands

Spectral Resolution and Feature Identification

- Generally surface features can be better distinguished from multiple narrow bands, than from a single wide band



Using the broad wavelength band 1, the features A and B cannot be differentiated

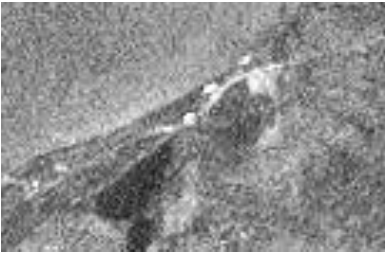
Spectral reflectance of A and B are different in the narrow bands 2 and 3, and hence can be differentiated

Spectral Resolution in Remote Sensing

- Different features are identified from the image by comparing their responses over different distinct spectral bands
- Broad classes, such as water and vegetation, can be easily separated using very broad wavelength ranges like visible and near-infrared
- For more specific classes viz., vegetation type, rock classification etc, much finer wavelength ranges and hence finer spectral resolution are required

Difference in the spectral responses of an area in different bands of Landsat TM image

TM Band 1
0.45–0.52 μm
(blue)



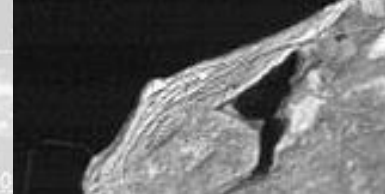
TM Band 2
0.52–0.60 μm
(green)



TM Band 3
0.63–0.69 μm
(red)



TM Band 4
0.76–0.90 μm
(near-infrared)



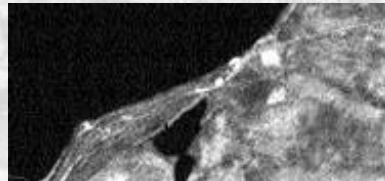
TM Band 5
1.55–1.75 μm
(mid-infrared)



TM Band 6
10.4–12.5 μm
(thermal-infrared)

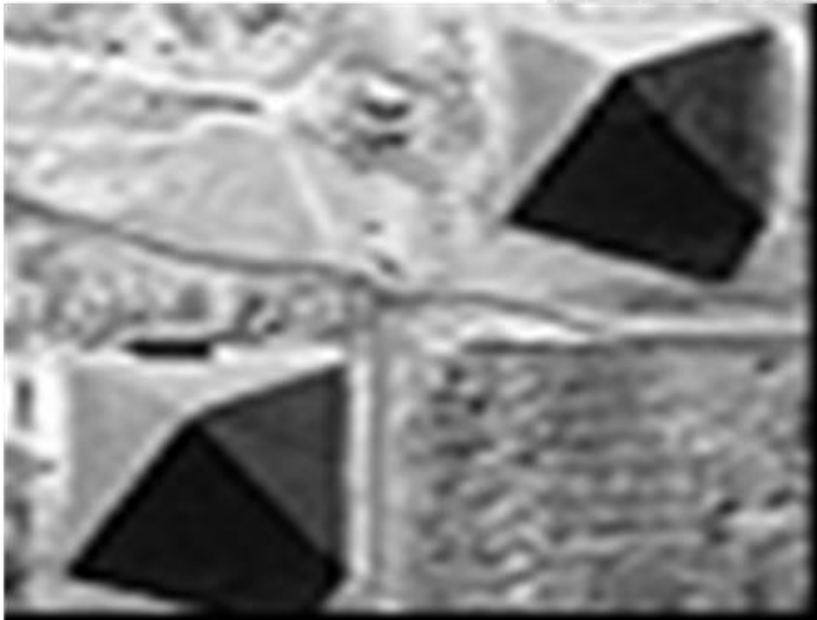


TM Band 7
2.08–2.35 μm
(mid-infrared)



Spectral Resolution...

Pan Image (Course)



Landsat TM RGB=543 (Fine)



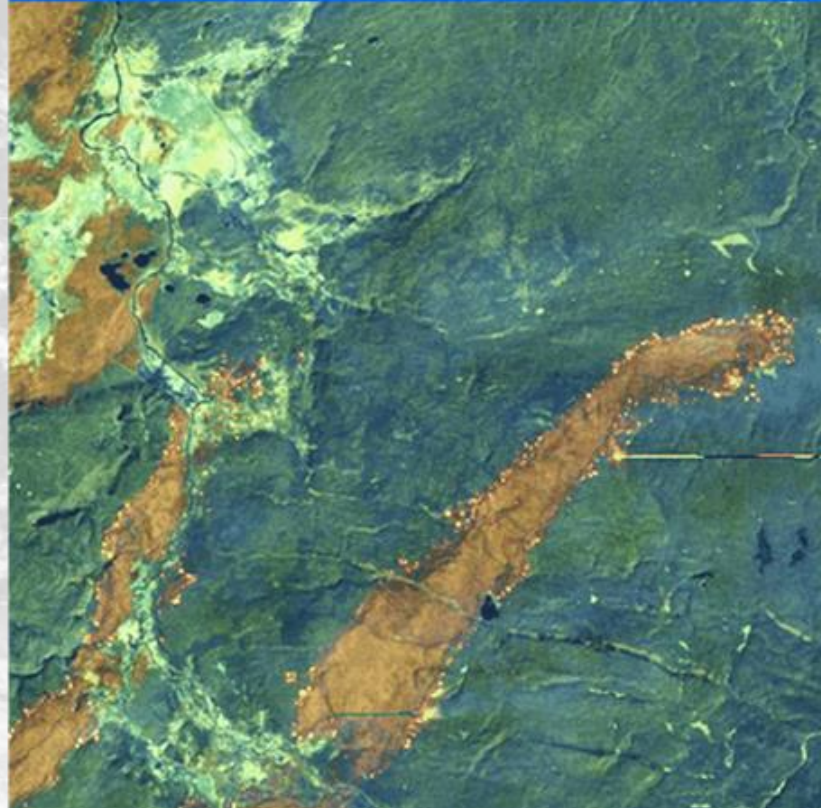
Spectral Resolution...

Forest Fire (Yellowstone NP)

Yellowstone NP, TCC (TM 321)



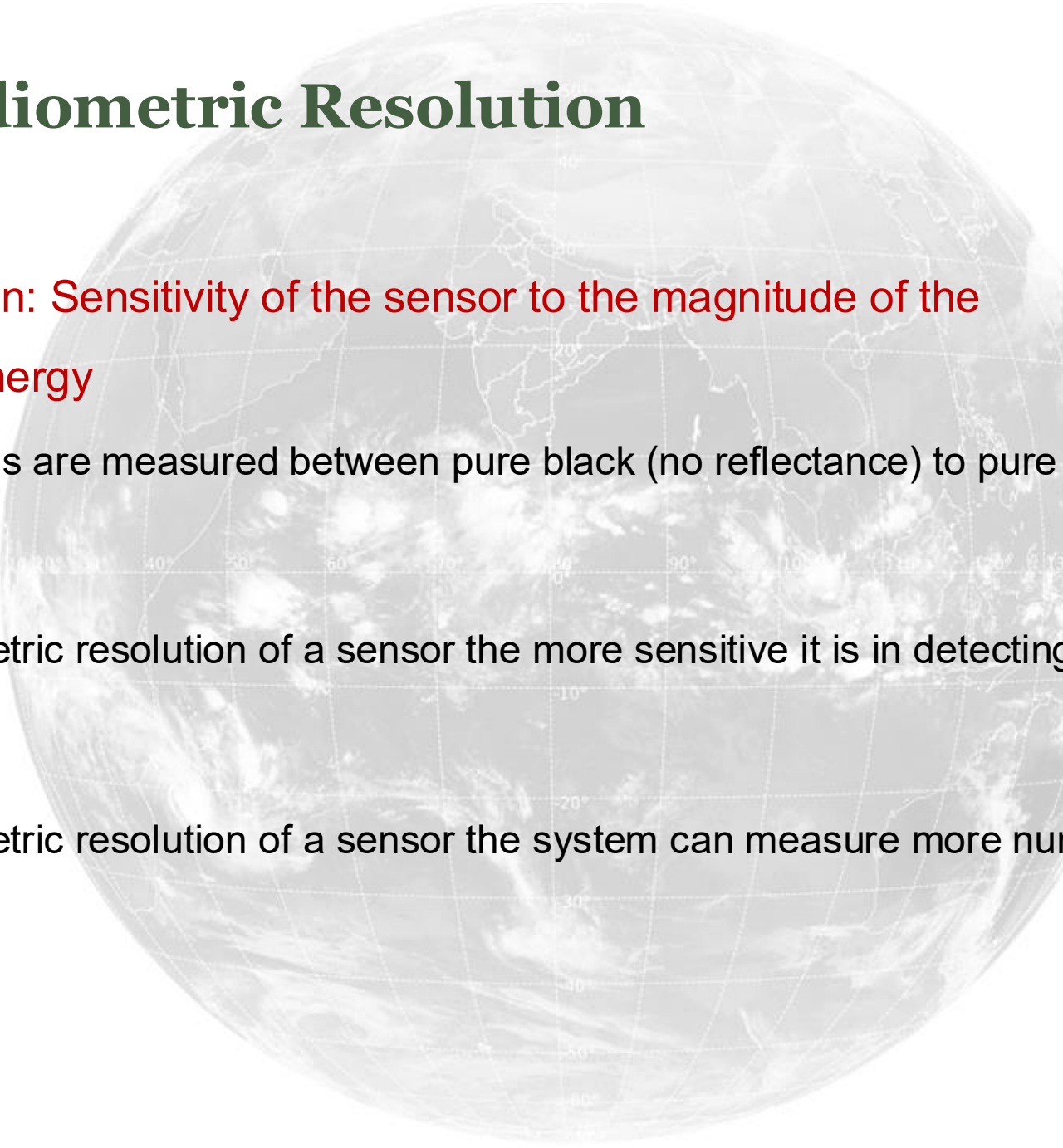
Yellowstone NP, FCC (TM 754)



Types of Resolution

- **4 types** of resolutions are defined for the remote sensing systems
 - ❖ Spatial resolution
 - ❖ Spectral resolution
 - ❖ Temporal resolution
 - ❖ Radiometric resolution

Radiometric Resolution



Radiometric resolution: Sensitivity of the sensor to the magnitude of the electromagnetic energy

- ❖ How many grey levels are measured between pure black (no reflectance) to pure white (maximum reflectance)
- ❖ The finer the radiometric resolution of a sensor the more sensitive it is in detecting small differences in the energy
- ❖ The finer the radiometric resolution of a sensor the system can measure more number of grey levels

Radiometric Resolution...

- Radiometric resolution is measured in **Bits**
 - Each bit records an exponent of power 2
- Maximum number of brightness levels available depends on the number of bits used in representing the recorded energy

Radiometric resolution and the corresponding brightness levels available

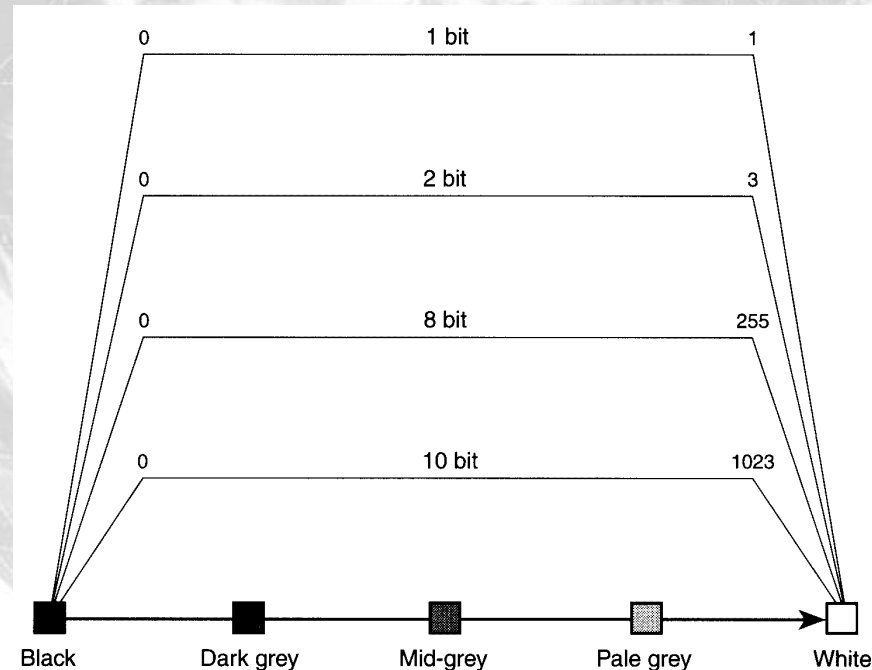
	Radiometric resolution	Number of levels	Example
Poor resolution →	1 bit	$2^1 - 2$ levels	
	7 bit	$2^7 - 128$ levels	IRS 1A & 1B
High resolution →	8 bit	$2^8 - 256$ levels	Landsat TM
	11 bit	$2^{11} - 2048$ levels	NOAA-AVHRR

Radiometric Resolution and Number of Grey Levels

- Tones in an image vary from black to white
- Black → Digital Number = 0 → No reflectance
- White → Digital Number is the maximum

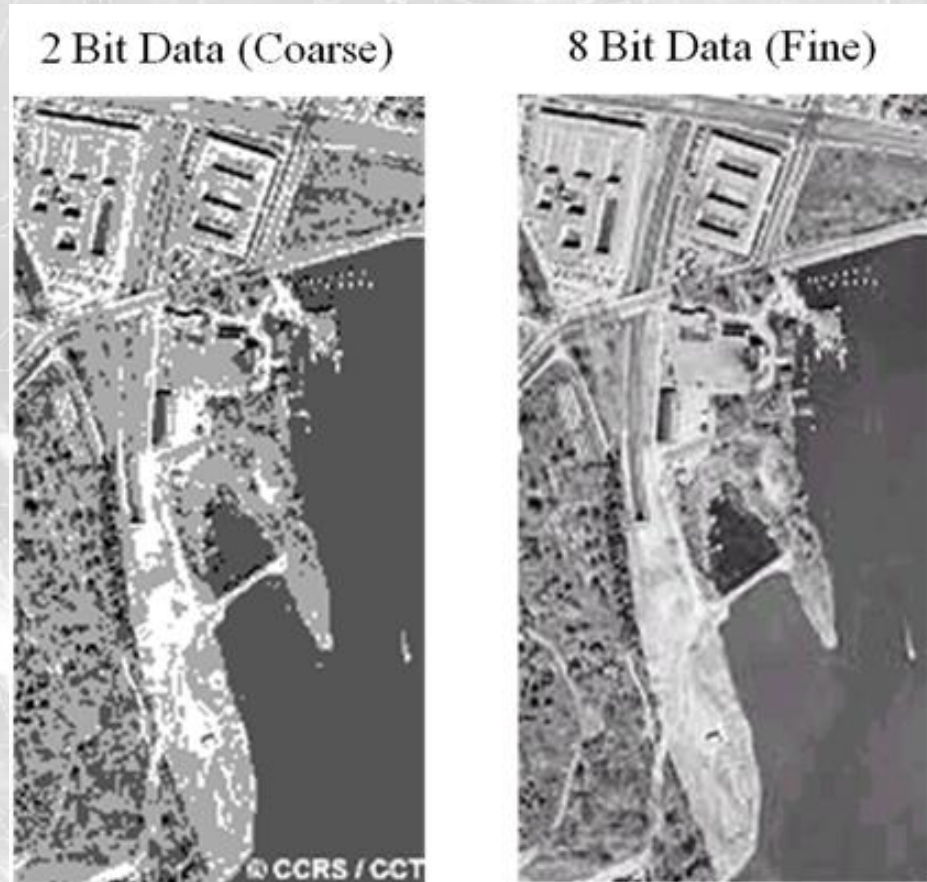
=1 for a 1-bit data

=255 for a 8-bit data



Radiometric Resolution and Level of Information

- Finer radiometric resolution
 - More the number of grey levels
 - More details can be captured in the image
- Finer radiometric resolution
 - Increases the data storage requirements



Radiometric Resolution and Digital Number

- Digital number (DN) depends on the number of brightness levels
 - Lower DN value in a coarse resolution image
 - Higher DN value in a fine resolution image

Example

- DNs recorded by the 3-bit system range from 0 to 7
- This range is equivalent to 0-63 for the 6 bit system

0	1	2	3	4	5	6	7	(3 bit)
0	9	18	27	36	45	54	63	(6 bit)

DN of 45 (6-bit) → 5 (3-bit)

To compare two images, their radiometric resolution should be the same

DIGITAL DATA

- ❑ Output from electronic sensors reaches the analyst as a set of digital values
- ❑ Each digital value is recorded as a series of binary (0 – 1) values known as *bits*
- ❑ Each *bit* records an exponent of power 2, with value of the exponent determined by position of the bit in the sequence
- ❑ Ex: A system is designed such that, each digital value is recorded using 7 bits
- ❑ Digital number 75 is recorded as 1001011
- ❑ Eight *bits* constitute one *byte*
- ❑ $1\text{ kb} = 2^{10}\text{ bytes} (= 1024\text{ bytes})$; $1\text{ Mb} = 2^{20}\text{ bytes}$; $1\text{ Gb} = 2^{30}\text{ bytes}$

DIGITAL DATA

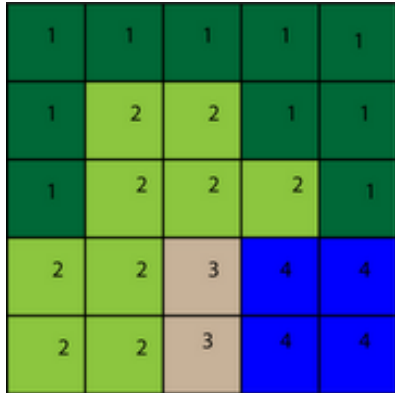
- ❑ The discrete value stored in each pixel in the digital format is called as DIGITAL NUMBER – DN
- ❑ The DN value for a pixel do not represent the true brightness (radiance) from the scene, but represents the relative brightness (scaled values)
- ❑ Number of brightness (DN) values within an image are determined by the number of *bits* available
- ❑ DNs are not comparable between two different scenes as DN value changes with *bits* available
- ❑ DNs are to be converted into radiance for comparison
- ❑ The sensitivity of a sensor (to represent brightness in terms of DN) has to be frequently calibrated with targets of known brightness

DATA FORMATS

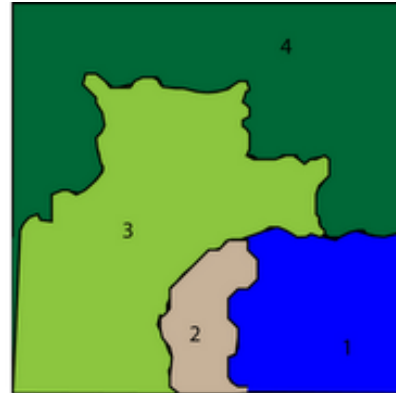
❑ Data in remote sensing can be stored in two formats

Raster Format:

- The image is treated as an array of values
- Each pixel is treated as a separate unit, and is designated using row and column index starting from upper left corner
- Each pixel has same dimension (defines resolution of image) and generally aligned in N-S direction
- Raster is faster – Manipulation of pixel values by image processing algorithms is easy (*since, pixels can be easily located*)
- In general, representation of actual ground objects (buildings; roads, vegetation;) are not so accurate in raster data system
- ***Vector Format*** uses polygonal patches and boundaries for representation and analysis

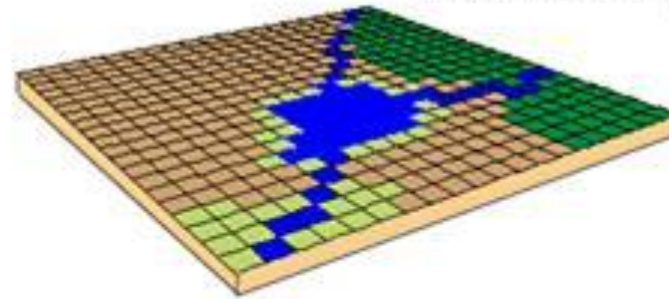


Values	Name	Count
1	Forest	10
2	Grass	9
3	Beach	2
4	Water	4

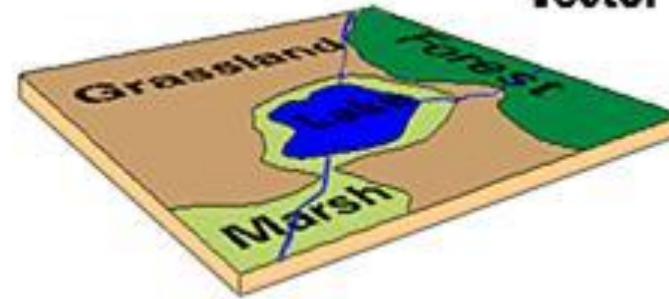


FID#	Name	value	Public?	Owner
1	Water	4	Yes	State
2	Beach	3	Yes	State
3	Grass	2	Yes	State
4	Forest	1	No	Warner

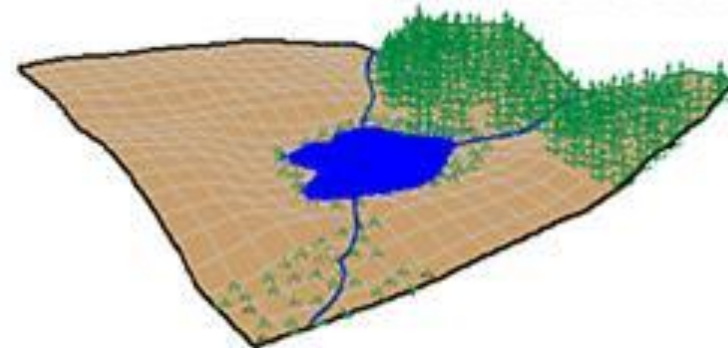
Raster / Image



Vector



Real World

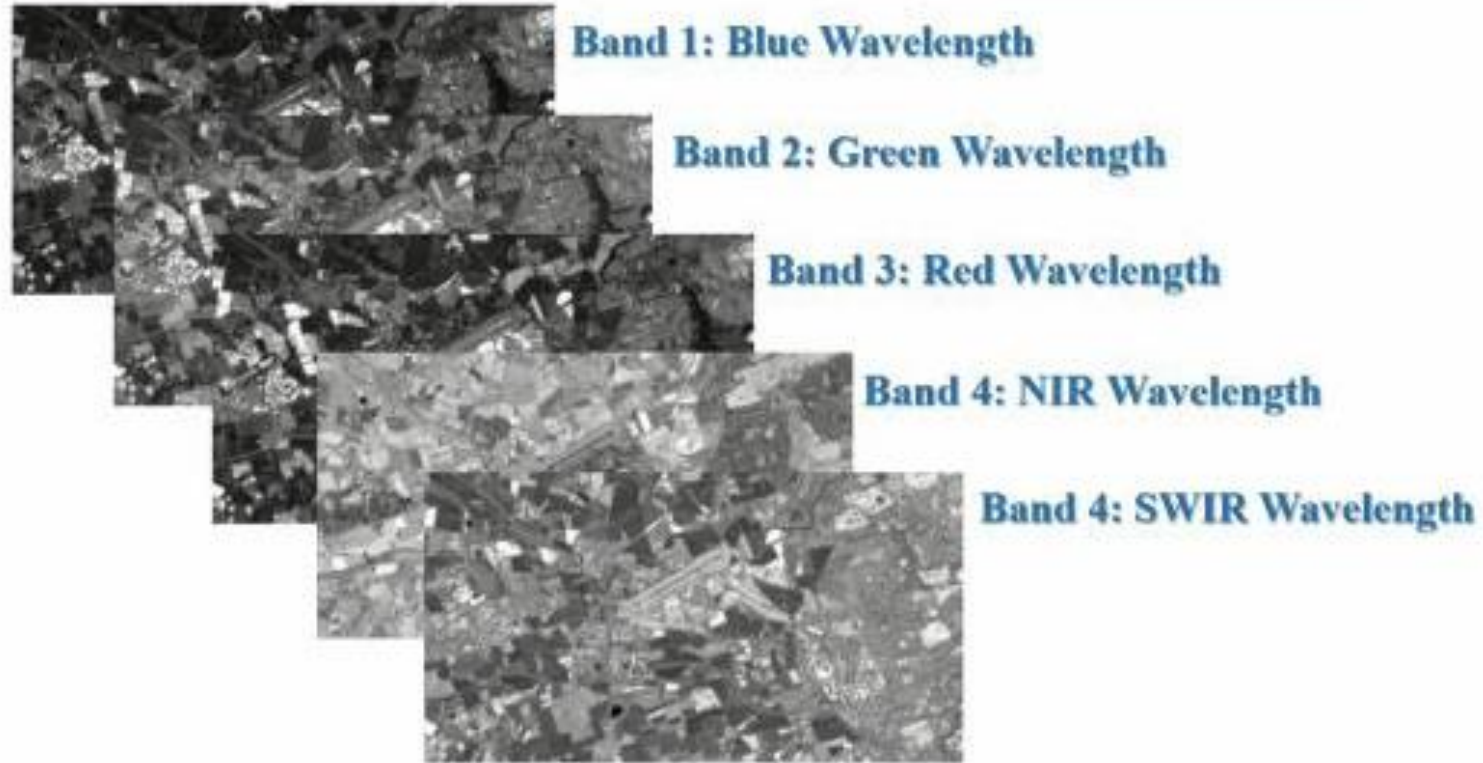


DIGITAL IMAGE FILE FORMATS

- ❖ **JPEG/JPEG 2000**
- ❖ **TIFF/GeoTIFF (Tagged Image File Format)**
- ❖ **MrSID**
- ❖ **PNG (Portable Network Graphics)**
- ❖ **HDF (Hierarchical Data Format)**
- ❖ **ASCII (American Standard Code for Information Interchange)**

How to save the acquired data in files?

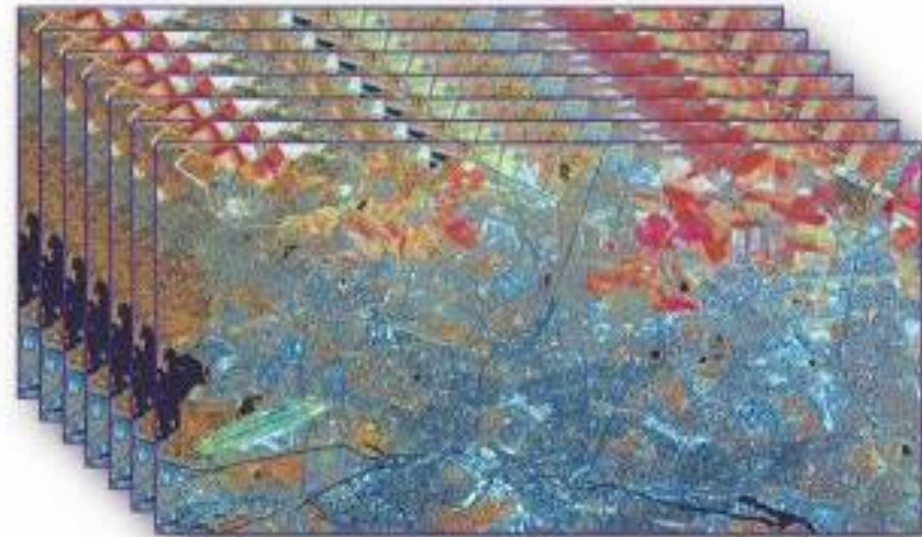
DIGITAL IMAGE FILE FORMATS



DIGITAL IMAGE DATA FORMATS

Formats in which image data is stored:

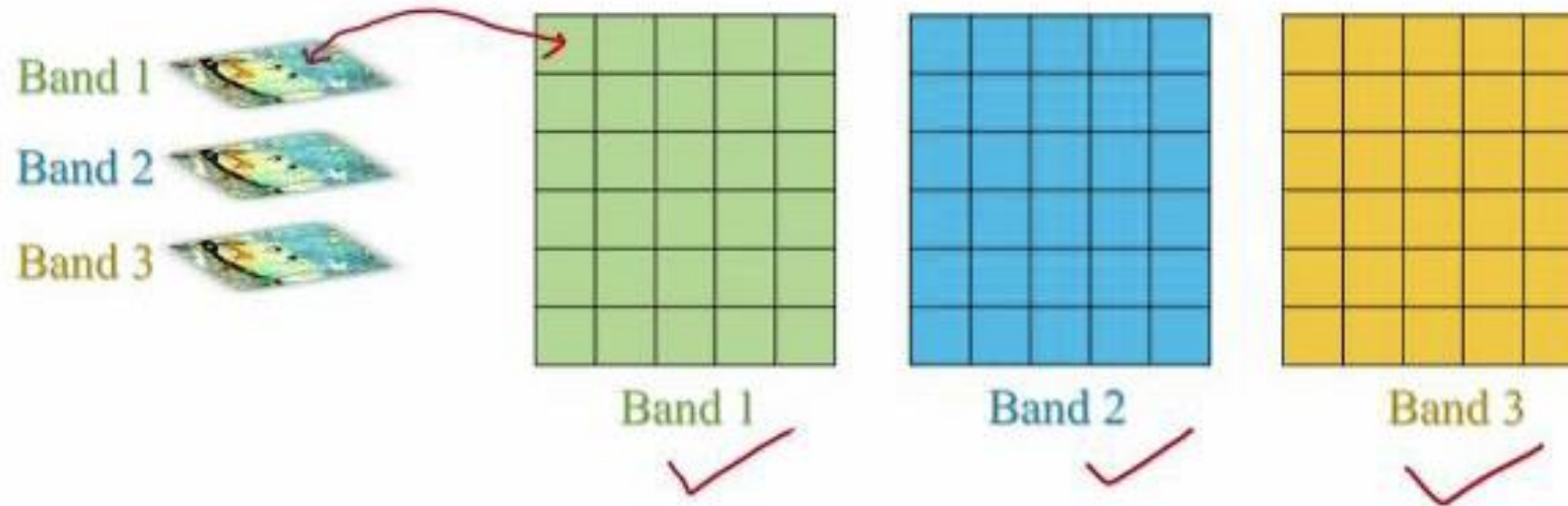
- ❖ Band Sequential (BSQ),
- ❖ Band Interleave by line (BIL),
- ❖ Band Interleave by pixel (BIP).



DIGITAL IMAGE DATA FORMATS

Band Sequential (BSQ):

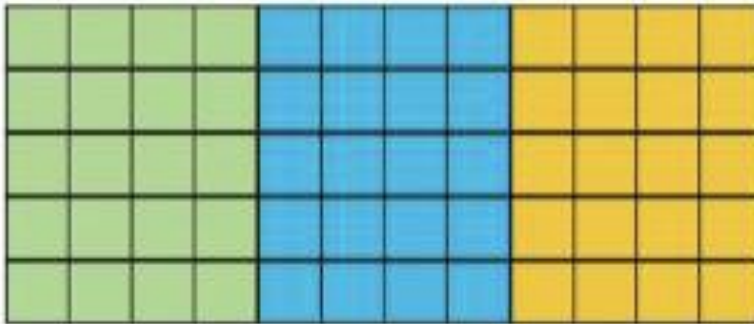
Data for a single band for entire scene is written in one file (*separate file for each band*)



DIGITAL IMAGE DATA FORMATS

Band Interleave by line (BIL):

Data for all the bands are written as line by line on the same file.



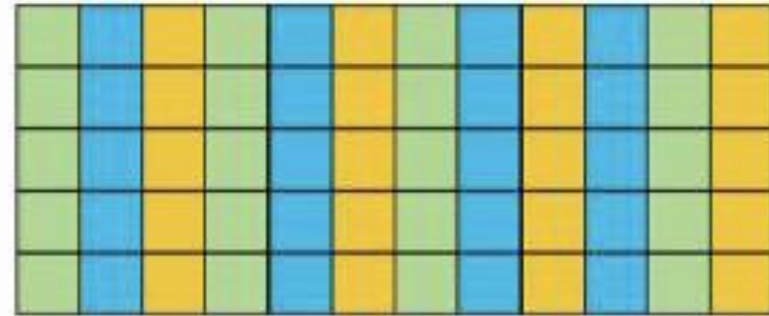
Band 1 

Band 2 

Band 3 

Band Interleave by Pixel (BIP):

Data for the pixels in all bands are written together.



Band 1 

Band 2 

Band 3 

DATA STORAGE IN RASTER FORMAT

- ❑ Consider an image in FOUR spectral channels (bands), which together can be visualized as FOUR superimposed images
- ❑ The pixel representing a feature will lie exactly on top of the same pixel (representing in all other bands) in superimposition mode
- ❑ There are THREE methods of storing such images

1) *Band Interleaved by Pixel (BIP)*

- ✓ Data are arranged in sequence values for

Row 1; Column 1; Band 1

Row 1; Column 1; Band 2

Row 1; Column 1; Band 3;

Row 1; Column 1; Band 4

- ✓ Next is for Row 1; Column 2; for all bands, and so on

DATA STORAGE IN RASTER FORMAT

- ❑ For a given pixel location, all the band values are stored sequentially
- ❑ Used for analyses of small area (limited pixels)
- ❑ Used for analyses in which, brightness value (DN) is queried or used to calculate another quantity

2) *Band Interleaved by Line (BIL)*

- ✓ Treats each line (row) data as a separate unit
- ✓ In sequence, the data is stored as:
 - Line (Row) 1 for Band 1;
 - Line (Row) 1 for Band 2;
 - Line (Row) 1 for Band 3;
 - Line (Row) 1 for Band 4;
- ✓ Next if for Line/Row 2 for all Bands, and so on

DATA STORAGE IN RASTER FORMAT

3) *Band Sequential Format (BSQ)*

- ✓ All data (pixels) specific to Band 1 are written first, followed by all data specific to Band 2, and so on
- ✓ For many (general) applications, BSQ format is convenient
- ✓ If areas smaller than the entire scene are to be examined, the analysis must read all four images before the sub area being identified
- ✓ The best data format depends on
 - a) application (use)
 - b) software / equipment available

Temporal Resolution

- Temporal resolution

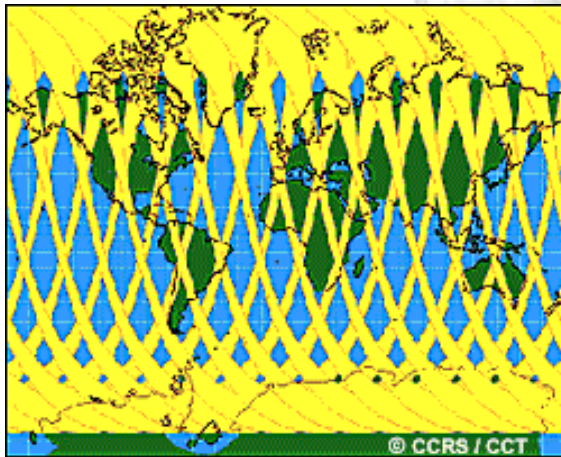
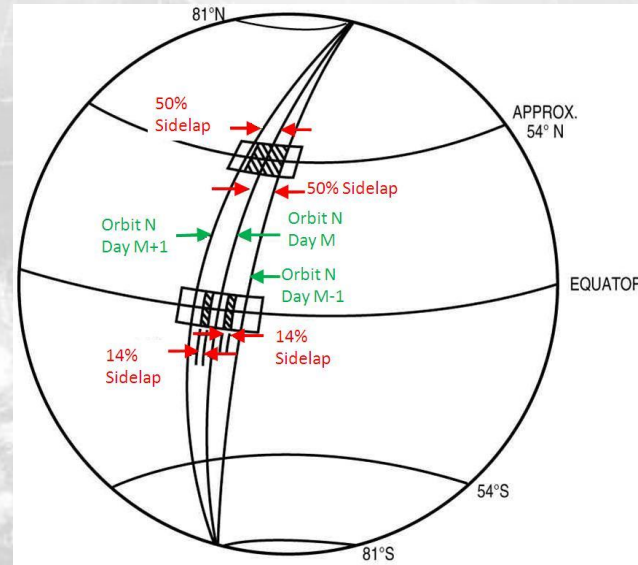
Number of times an object is sampled
or

How often data are obtained for the same area

- ❖ The absolute temporal resolution of a remote sensing system to image the same area at the same viewing angle a second time is equal to the **repeat cycle of a satellite**.
- ❖ The repeat cycle of a near polar orbiting satellite is usually several days
Example: 24 days for IRS-1C and Resourcesat-2, 18 days for Landsat, 14days for IKONOS
- ❖ Actual temporal resolution (or revisit period) of a sensor depends on
 - The satellite/sensor capabilities
 - Swath overlap and Latitude

Swath Overlap and Latitude

- ❖ Sidelap in the swaths of the adjacent orbits increases the frequency of imaging
- ❖ Sidelap increases with latitude, increasing the frequency of images available for the polar region



Paths of a Typical Near-Polar Satellite

Towards the polar region, satellite orbits come closer to each other. More frequent images are available for the polar region

Satellite Capabilities and Temporal Resolution

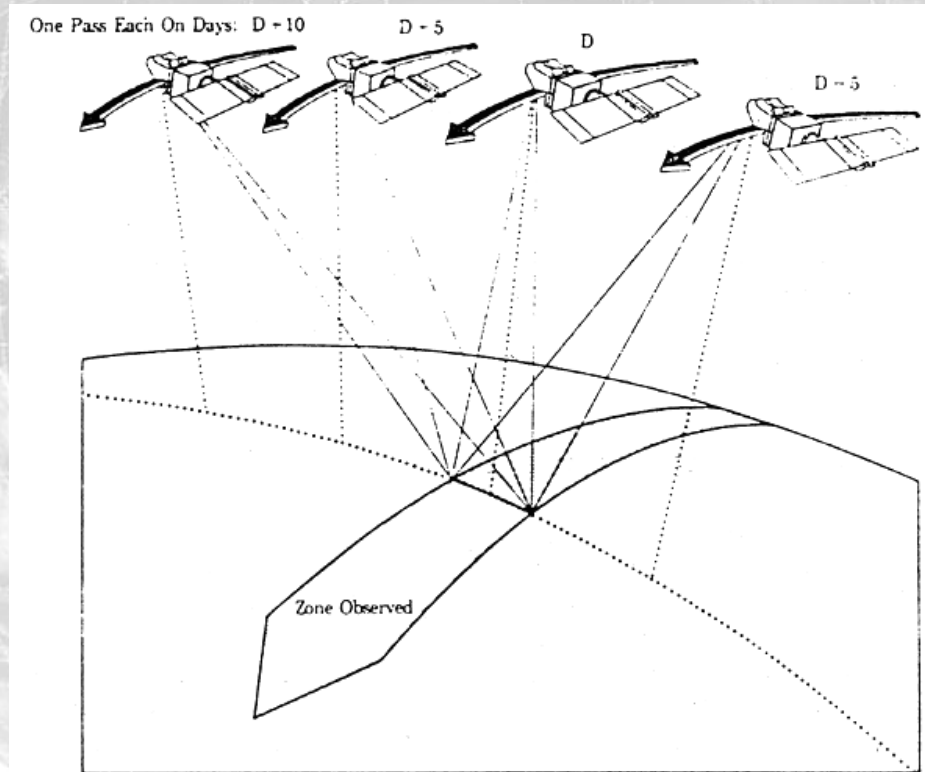
- More frequent imaging is possible by off-nadir viewing capabilities

Example : IKONOS

Sensor characteristics: Pointable optics

Repeat cycle : 14 days

Revisit period : 1-3 days



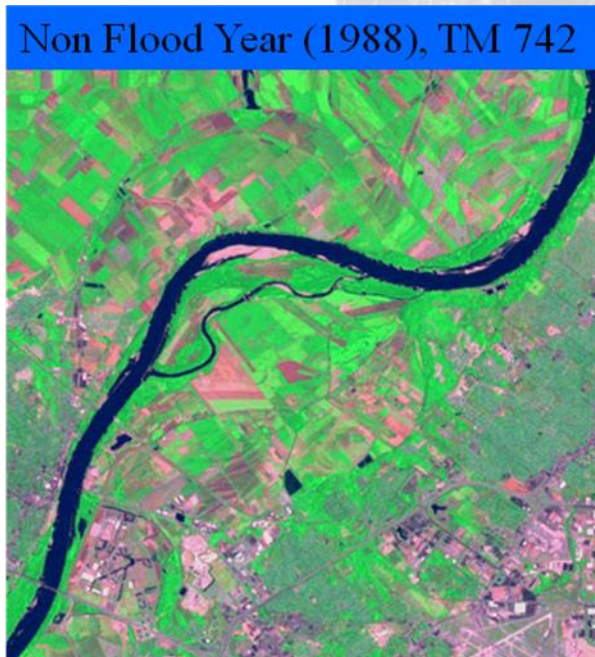
Importance of Temporal Resolution

- Images at different time periods show the variation in the spectral characteristics of different features over time
- Applications
 - Land use/ land cover classification
 - Temporal variation in land use / land cover
 - Monitoring of a dynamic events like
 - ✦ Cyclone
 - ✦ Flood
 - ✦ Volcano
 - ✦ Earthquake

Flood Studies

- Satellite images before and after the flood event help to identify the aerial extent of the flood during the progress and recession of a flood event

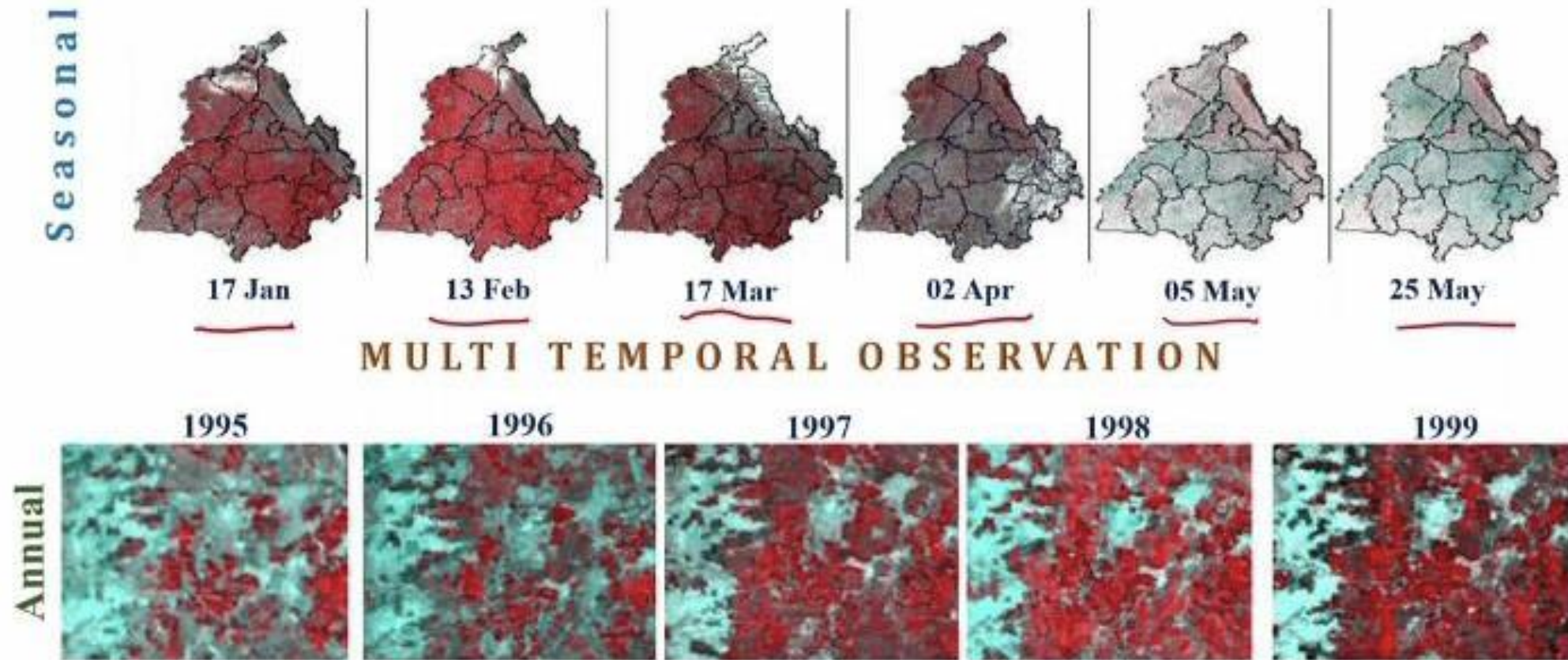
Landsat TM images of the Mississippi River taken during a normal period and during the great flood of 1993



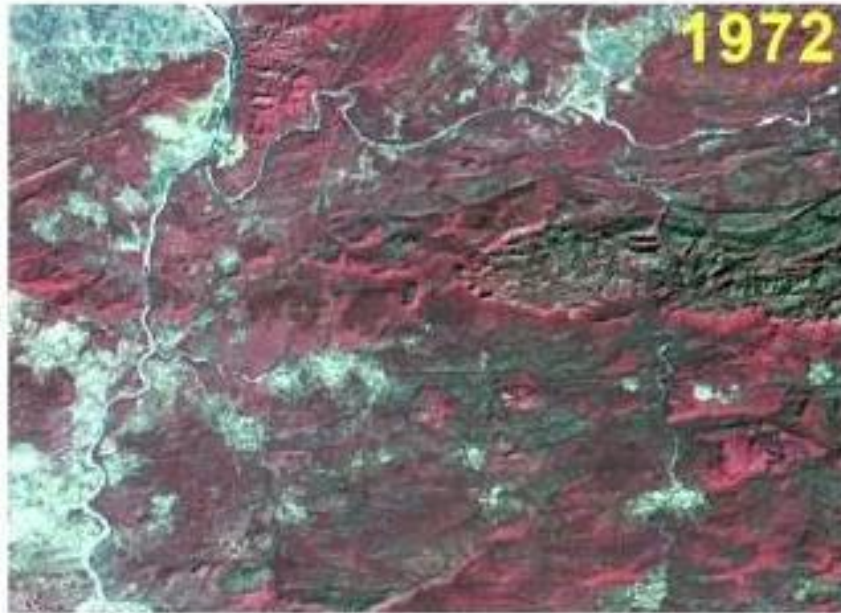
Land Use/ Land Cover Classification

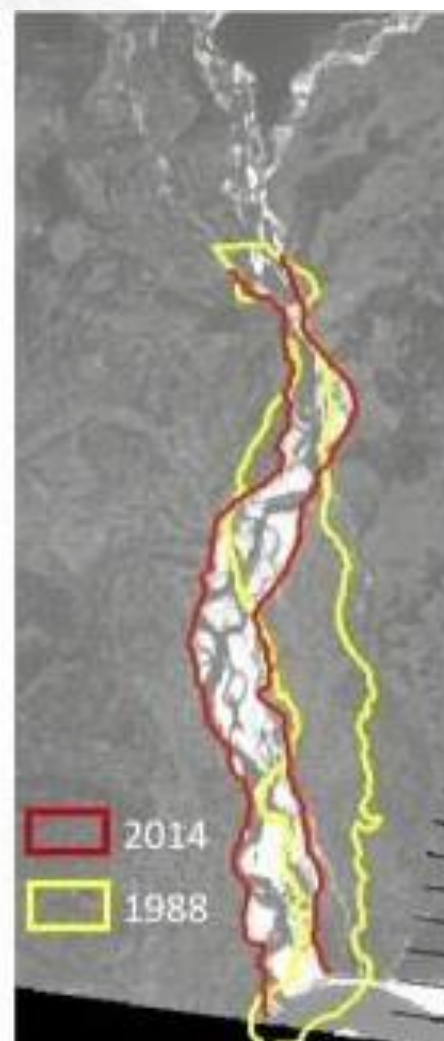
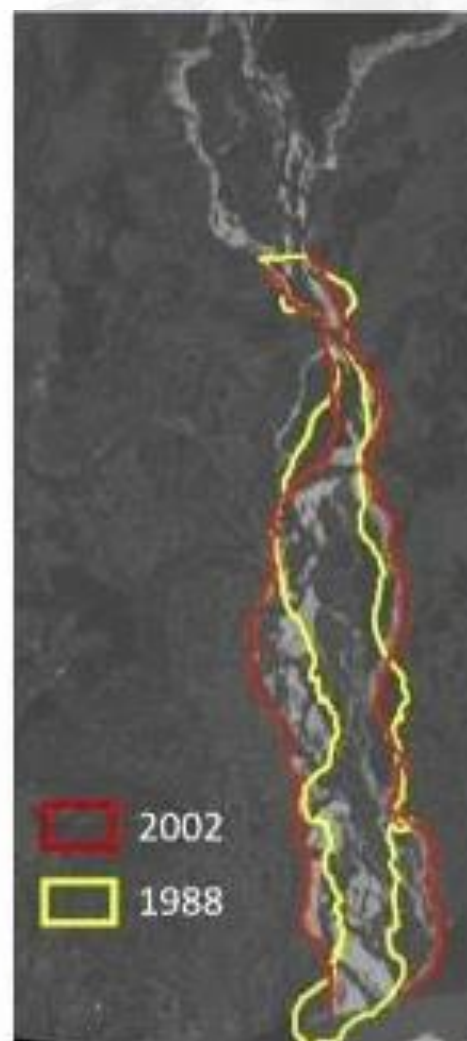
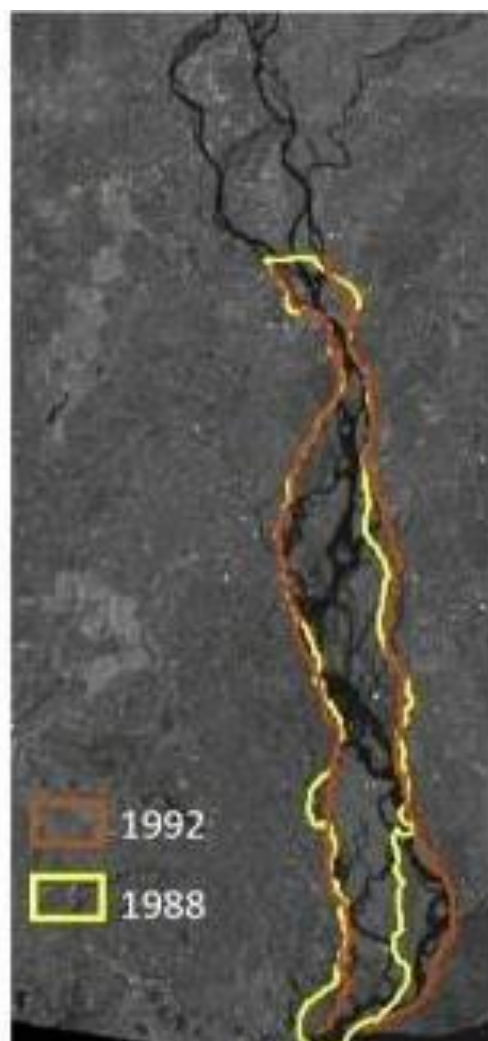
- Temporal variation in the spectral signature can be estimated
- Presence of features over time can be identified
- Continuous change in the vegetation characteristics can be monitored
 - Used to classify the crop types viz., perennial crops, long or short duration crops

Crop Monitoring :

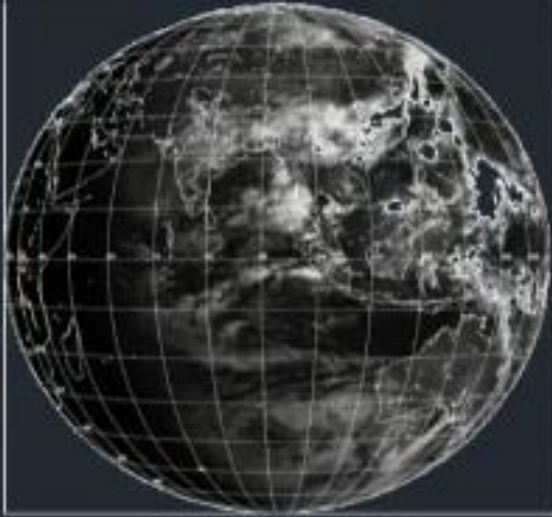


Long term

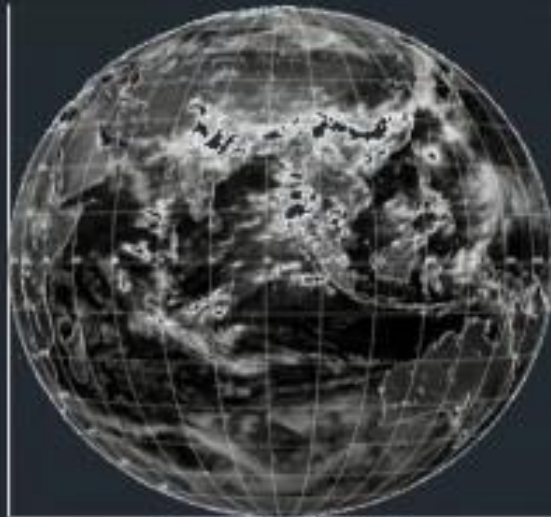




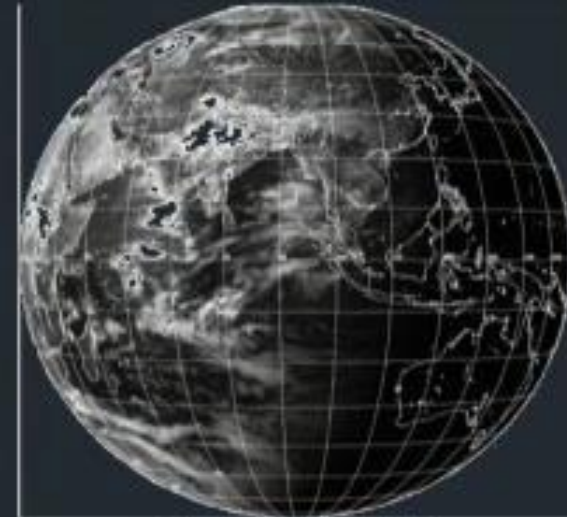
INSAT 3A - Visible image



Aug 22, 2012 02:00



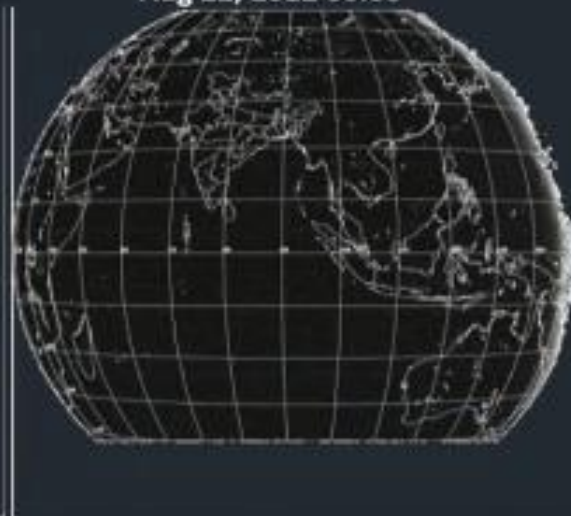
Aug 22, 2012 06:00



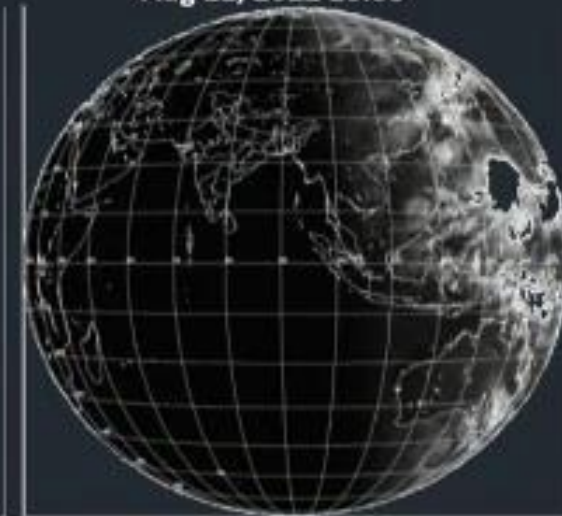
Aug 22, 2012 10:00



Aug 22, 2012 16:00



Aug 22, 2012 20:00



Aug 22, 2012 23:00

Signal-to-Noise Ratio

- ❖ Signal-to-noise ratio (SNR) depends on strength of signal and the noise of the system
 - Signal (say reflectance)
 - Noise from aberrations in the electronics, moving parts or defects in the scanning system
- ❖ Higher the SNR → Differentiation of the noise from the actual signals is easier
- ❖ Finer spatial, spectral and radiometric resolutions of a system may decrease the SNR to such an extent that the data may not be reliable
 - Higher spectral and spatial resolution reduces the energy (signal strength) → reduces the SNR
 - Finer radiometric resolution → larger number of grey levels
 - If the difference in the energy level between the two levels is less than the noise, reliability of the recorded grey level diminishes.

Trade-off Between Resolutions

Fine spatial resolution → small IFOV → less energy

- Difficult to detect fine energy differences → Poor radiometric resolution
- Poor spectral resolution

Narrow spectral bands → High spectral resolution → Less energy

- Difficult to detect fine energy differences → Poor radiometric resolution
- Poor spatial resolution

Wide spectral band → Poor spectral resolution → more reflected energy

- Good spatial resolution
- Good radiometric resolution

These three types of resolutions must be balanced against the desired capabilities and objectives of the sensor

SPECTRAL INDICES

- Spectral indices → Combinations of surface reflectance at two or more wavelengths that indicate relative abundance of features of interest
- Indices can be based on 'difference' or 'ratio' or 'a combination' of reflectance in various bands
- Spectral indices are mainly used in vegetation, and geological applications

Vegetation Spectral Indices

Combining the reflectance in more than one spectral bands to create a numeric quantity. Some of the objectives are:

- To improve sensitivity to some of the biophysical properties of the plant.
- To reduce the effects of (at least a few)
 - Sun angle
 - Sensor viewing angle
 - Atmosphere
 - Canopy background effect
 - Soil effect
 - Topography effect

IMAGE RATIOS

- It is possible to *divide the digital numbers/reflectance* of one image band by those of another image band to create a third image.
- Ratio images may be used to remove the influence of light and shadow on a ridge due to the sun angle.
- It is also possible to *calculate certain indices* which can enhance vegetation or geology
- **Vegetation: NIR/Red (Healthy vegetation → Large ratio; Stressed Vegetation → Small ratio)**

Sensor	Image Ratio	EM Spectrum	Application
Landsat TM	Bands 3/2	red/green	Soils
Landsat TM	Bands 4/3	PhotoIR/red	Biomass
Landsat TM	Bands 7/5	SWIR/NIR	Clay Minerals/Rock Alteration

Normalized Difference Vegetation Index (NDVI), a commonly used vegetation index which uses the red and infrared bands of the EM spectrum.

SPECTRAL INDICES

Common Vegetation Indices

1) Normalized Difference Vegetation Index (NDVI) →

$$NDVI = \frac{(\rho_{NIR} - \rho_{RED})}{(\rho_{NIR} + \rho_{RED})}$$

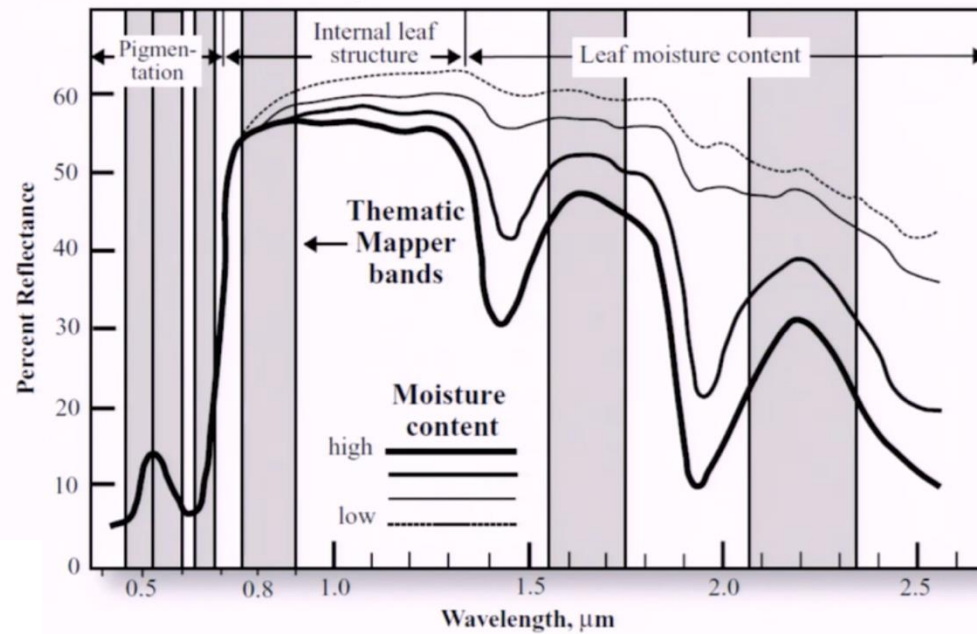
- Range -1 to + 1
- This index is a measure of healthy, green vegetation
- Uses the combinations of highest absorption and reflectance regions of Chlorophyll
- It can saturate in dense vegetation conditions when LAI becomes high

2) Green-NDVI → $GNDVI = \frac{(\rho_{NIR} - \rho_{GREEN})}{(\rho_{NIR} + \rho_{GREEN})}$

- This index is more sensitive to chlorophyll concentration than NDVI

Observing vegetation from Landsat TM

Reflectance of a Sycamore Leaf at Different Moisture Conditions



Normalized Difference Water/Moisture Index (NDWI/NDMI)

$$NDWI = \frac{\rho_{NIR} - \rho_{SWIR}}{\rho_{NIR} + \rho_{SWIR}}$$

Correlated with vegetation liquid water content
(Gao, 1996)

Another definition of NDWI

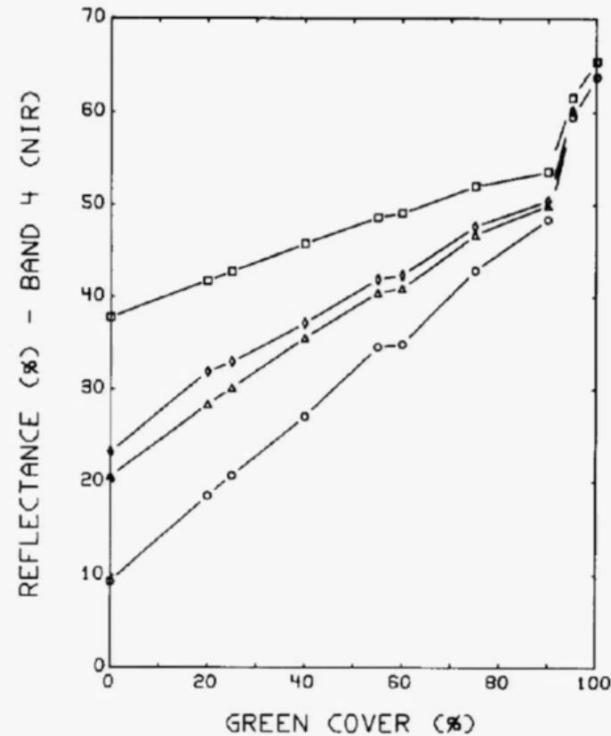
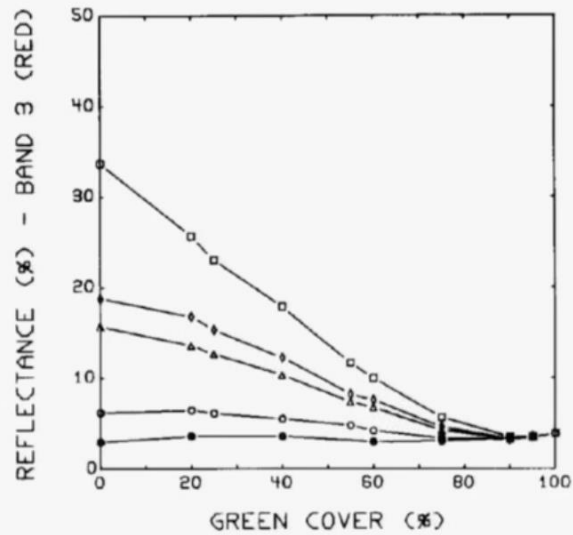
$$NDWI = \frac{\rho_{Green} - \rho_{NIR}}{\rho_{Green} + \rho_{NIR}} \quad (\text{McFeeters, 1996}) \quad \text{GreenNDVI}$$

$$MNDWI = \frac{\rho_{Green} - \rho_{SWIR}}{\rho_{Green} + \rho_{SWIR}} \quad (\text{Xu, 2006})$$

For identifying open water bodies

Often confused with NDWI/NDMI for liquid vegetation water content

Factors affecting NDVI other than vegetation! – Soil background

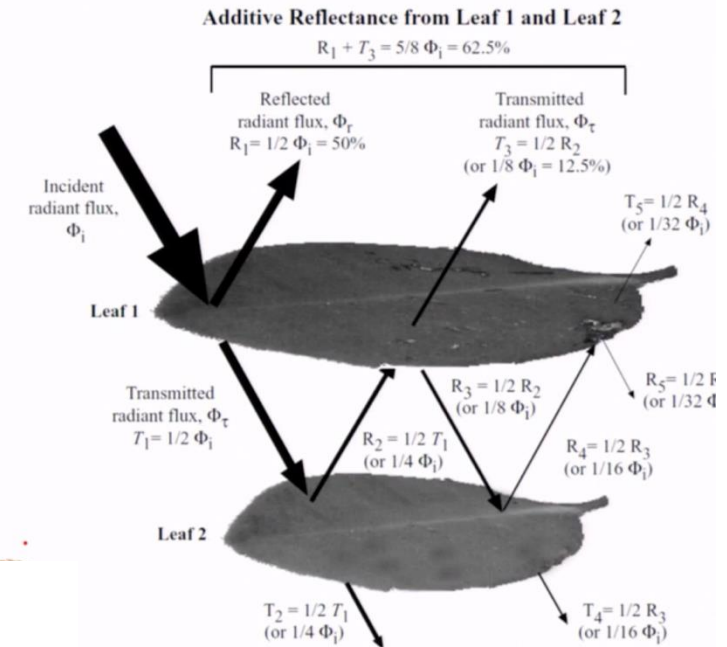


Soil Adjusted Vegetation Index (SAVI) →

$$SAVI = \frac{1.5 \times (\rho_{NIR} - \rho_{RED})}{(\rho_{NIR} + \rho_{RED} + L)}$$

- This index is similar to NDVI, but it suppresses the effects of soil pixels
- Uses a canopy background adjustment factor, L , (~ 0.5) which is a function of vegetation density
- This index is best used in areas with relatively sparse vegetation where soil is visible through the canopy

Leaf additive reflectance

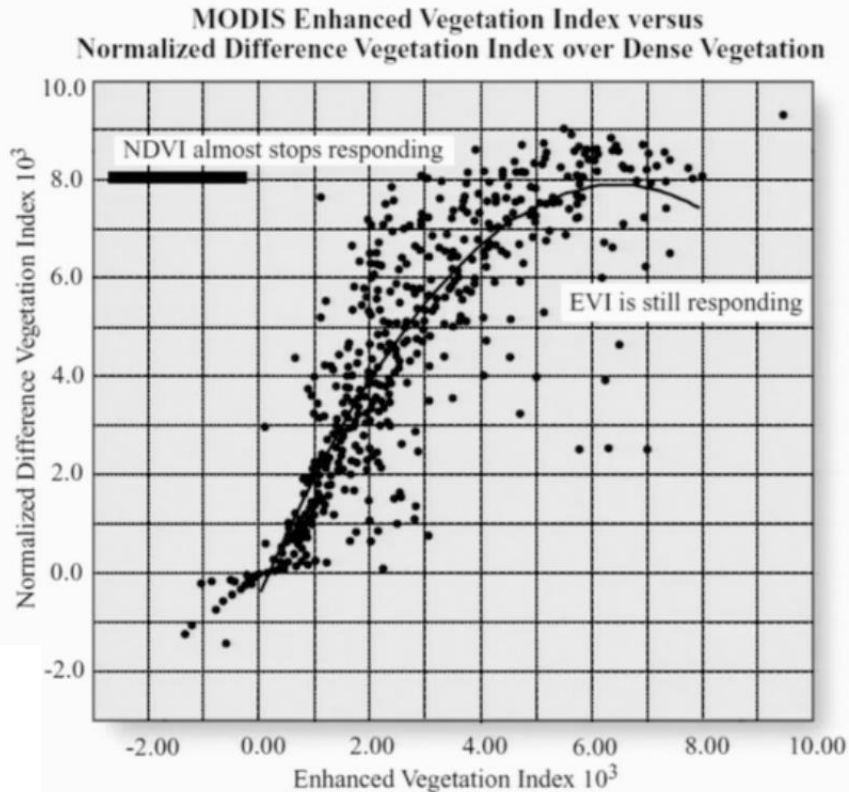


Presence of multiple leaves will change the reflectance observed.

- $L = 1$ (Low vegetation)
- $L = 0.5$
- $L = 0.25$ (High vegetation)

Enhanced Vegetation Index (EVI)

$$EVI = G \frac{\rho_{nir}^* - \rho_{red}^*}{\rho_{nir}^* + C_1 \rho_{red}^* - C_2 \rho_{blue}^* + L} (1 + L).$$



$L = 1$

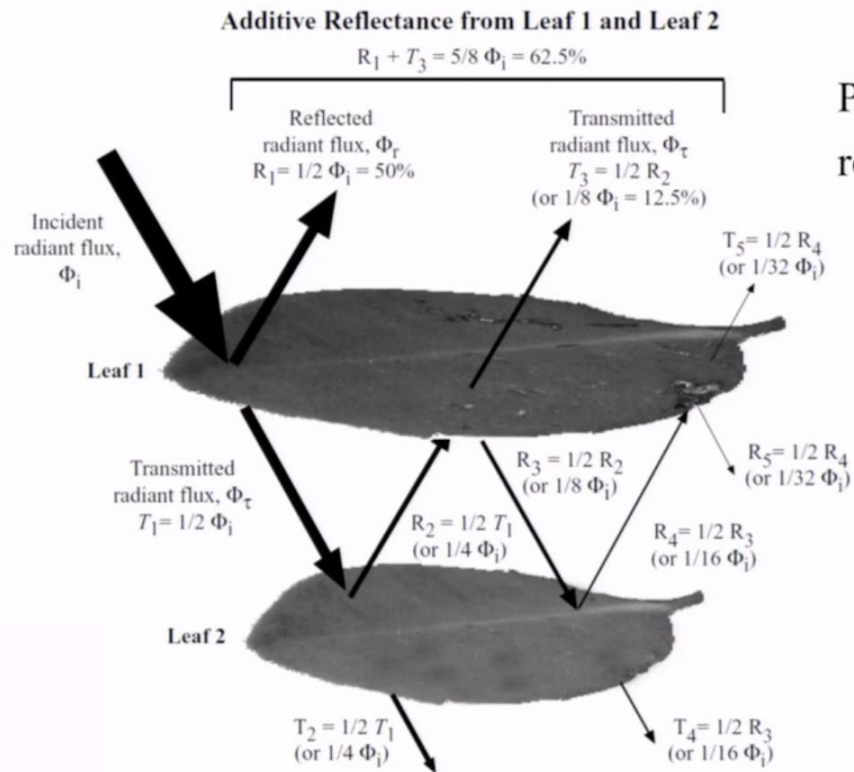
$C_1 = 6$

$C_2 = 7.5$

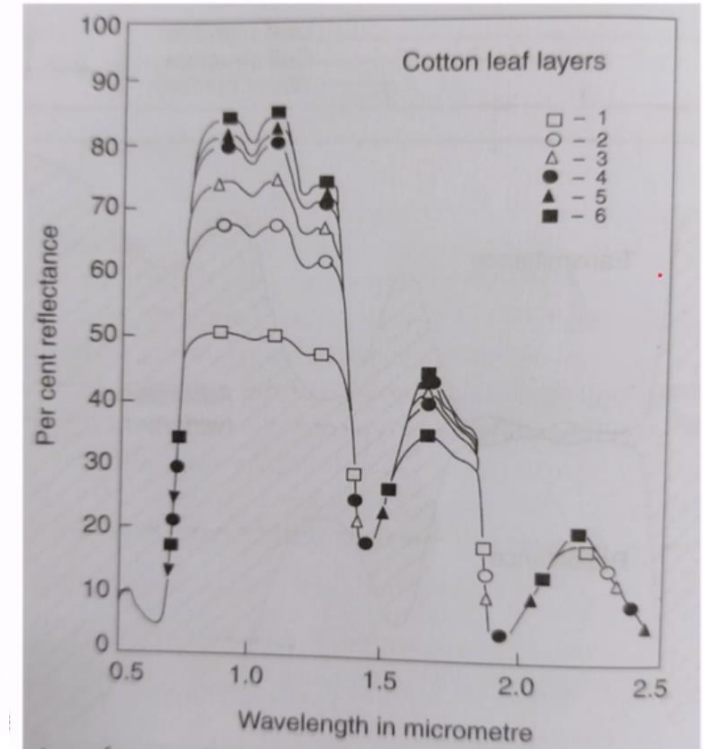
$G = 2.5$ (gain factor)

- Used in optimizing the vegetation signal in areas of high LAI
- Useful in high LAI regions where NDVI may saturate
- Uses the blue reflectance region to correct for soil background signals and to reduce atmospheric influences, including aerosol scattering

Leaf additive reflectance



Presence of multiple leaves will change the reflectance observed.



Increase in reflectance is clearly observed the **NIR** band.

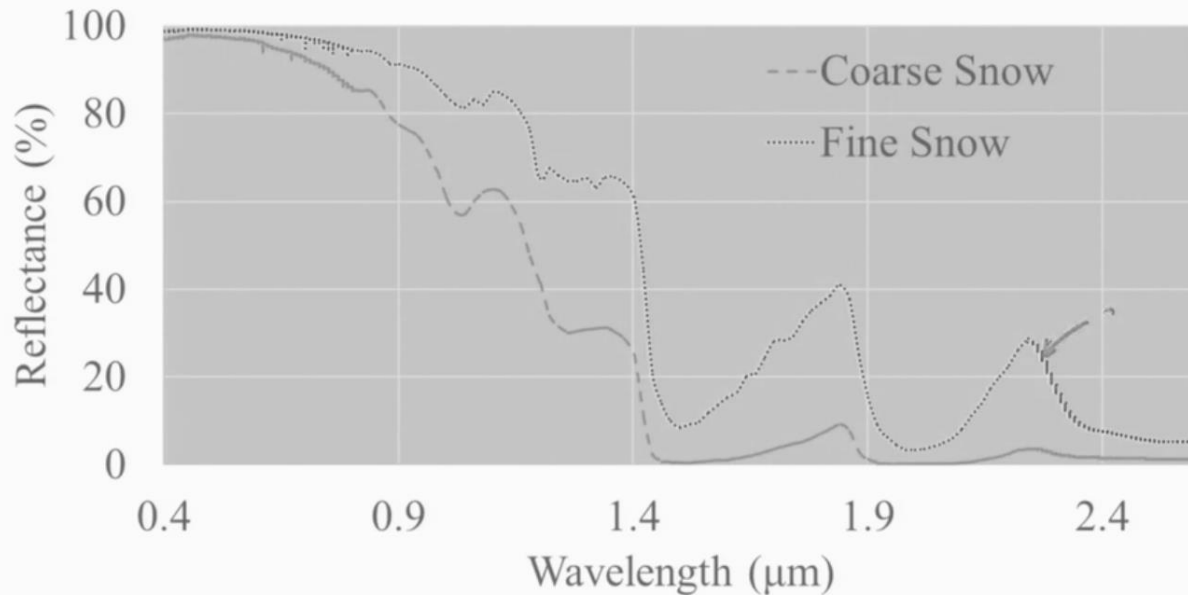
Leaf Area Index (LAI) $\rightarrow LAI = (3.618 * EVI - 0.118)$

- This index is used to estimate foliage cover and to forecast crop growth and yield

Spectral properties of Snow

- Snow is a collection of ice grains and air containing soluble and particulate impurities.
- Snow exhibits wide variation in grain size, density, depth etc.
- Snow has the highest reflectance in visible region compared to any other material.
- Snow shows marked variation in reflectance due to metamorphism, melting and other properties.

Snow reflectance: Wavelength



Reflectance is much higher in visible bands and reduces significantly in IR bands

Normalized Difference Snow Index (NDSI)

$$NDSI = \frac{\rho_{green} - \rho_{SWIR}}{\rho_{green} + \rho_{SWIR}}$$

$$\text{Landsat TM NDSI} = \frac{(\text{TM } 2 - \text{TM } 5)}{(\text{TM } 2 + \text{TM } 5)}$$

$$\text{MODIS NDSI} = \frac{(\text{MODIS } 4 - \text{MODIS } 6)}{(\text{MODIS } 4 + \text{MODIS } 6)}$$

SPECTRAL INDICES

For a detailed list of spectral indices used in RS, refer to

<https://www.nv5geospatialsoftware.com/docs/SpectralIndices.html>